Structure Over Axioms: From Inherent Mathematical Constraints to Three-Dimensional Bases and Omnidimensional Reconstruction

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The preceding physical research project, the **Space Interlayer Conjecture / Phase Manifestation Mechanics** (abbreviated as **SIC / PMM**), exposed several blind spots in modern mathematical systems concerning discreteness, numerical definitions, and mathematical representation. Based on these issues, this paper proposes and develops the central principle of **Structure Over Axioms**.

This paper argues that axioms should not be regarded as the absolute starting point of mathematical deduction. Rather, they should be understood as expressions of underlying mathematical structures under specific conditions and dimensional constraints. The definability of a mathematical object cannot be established independently of its generative premises, conditions of existence, and structural boundaries. To examine this principle concretely, this paper first analyzes cases such as $\tan(90^\circ)$, investigating whether a numerical value remains definable when its original prerequisite structure disappears. It then addresses issues involving the discrete and the continuous, infinity and boundaries, structural extension, and the criteria by which an extension may be determined to be consistent and well-defined.

Within the framework of **Structure Over Axioms**, this paper revisits the three major crises in the foundations of mathematics and investigates whether their underlying mechanisms can be described through a common foundational structure. Subsequently, the paper uses the specialized equations of **PMM** as a concrete construction case. By modifying key extension conditions, it establishes a new mathematical structure and further discovers that this structure can form a **three-dimensional basis** possessing properties of a generative or mother structure. On this basis, corresponding operators, operational rules, and structural representations are defined.

After the gradual formation of this mathematical structure, the paper further identifies possible extensions into several fields that were not initially presupposed as research objectives. These include the structural reformulation and dimensional reduction of the seven Millennium Prize Problems, the geometric necessity of probability, and metacognition in artificial intelligence and new AGI architectures. These topics were not the original goals of the study; rather, they emerged as subsequent possibilities through the processes of structural reconstruction and extension.

Keywords: Structure Over Axioms, Inherent Mathematical Constraints, Three-Dimensional Base, Foundational Mathematics & Axiomatic Logic, Mathematical Crises & Structural Foundations, Higher-Dimensional Geometry & Topology, Phase Manifestation Mechanics, Structural Approach to Millennium Prize Problems, Probabilistic High-Dimensional Geometry, Formalization of AGI Metacognitive Structures

STRUCTURE OVER AXIOMS

Throughout the development of modern mathematics, formalism and axiomatic methods—such as the Hilbert Program and Zermelo–Fraenkel set theory with the Axiom of Choice (ZFC)—have achieved remarkable success, advancing mathematics toward a highly abstract, rigorous, and productive framework. However, when viewed from the perspective of a cross-dimensional and holographic omnidimensional framework, a central crisis within modern mathematics appears to manifest in two fundamental dimensions:

1. Over-Expansion of Axioms

Whenever a mathematical system encounters a new singularity, an unresolved contradiction, or an independence result—such as the Continuum Hypothesis—the conventional response is often to introduce an additional axiom to address the problem.

This continual expansion causes contemporary mathematical systems to become increasingly cumbersome, fragmented, and disconnected from a unified organic generative principle. Mathematics thereby risks entering an endless cycle:

> New axioms are introduced to solve problems, and the system is expanded further to support those new axioms.

2. Isomorphism of Infinity

Whether in Cantor's hierarchy of infinite cardinalities in set theory, the distribution of prime numbers in number theory, or the deep correspondences found in topology and algebraic geometry, different branches of mathematics frequently reveal highly consistent underlying patterns when addressing **infinity** and **limits**.

This recurring structural similarity suggests that many of the profound difficulties and apparent incommensurabilities encountered in mathematics may originate not from intrinsic contradictions within the underlying structures, but from the failure or limitations of the mathematical tools used to describe them.

1. RESEARCH ORIGIN: MATHEMATICAL BLIND SPOTS REVEALED BY THE PHYSICS STUDY

The preceding physical research, the **Space Interlayer Conjecture / Phase Manifestation Mechanics** (abbreviated as **SIC / PMM**), revealed five major blind spots, among which the central mathematical bottleneck concerns the constraints imposed by discrete structures and positional numeral systems.

The existing mathematical framework relies heavily on discretization and specific numerical representations based on positional numeral systems. When such representations are used to derive continuous and higher-dimensional physical realities, the complexity of successive operations inevitably introduces and accumulates structural errors. During the development of the preceding physics paper, it was therefore necessary to compromise with the existing mathematical framework and employ conventional mathematical language to connect the proposed theory with established physical equations. However, this transitional approach, in turn, exposed the fundamental limitations of the existing mathematical tools.

Accordingly, the starting point of the present paper is to move beyond this transitional mode of representation and return directly to the underlying mathematical structure itself. The objective is to establish a new mathematical framework capable of providing a precise description of **SIC / PMM**, simplifying the representation of higher-dimensional operators, and fundamentally eliminating structural distortions arising from the limitations of conventional mathematical representations.

1.1 Discrete Structures and the Limitations of Mathematical Definitions

Structural Distortion of Positional Numeral Systems:

The existing real-number system and algebraic operations rely heavily on specific **positional numeral systems (base representations)**, while often overlooking the fact that the choice of such a representation framework itself implicitly establishes a reference for distinguishing between **rational and irrational numbers**.

Positional numeral systems fundamentally originate from the practical convenience of human counting. When continuous, high-dimensional phase structures are forcibly embedded into discrete grids defined by finite-base representations, the resulting infinite non-repeating decimal expansions or truncation errors may be misinterpreted as intrinsic **randomness** or **singularity** of the numerical values themselves.

From the perspective of structural analysis, however, such phenomena may instead reflect limitations imposed by the representational framework rather than properties inherent to the underlying structure.

1.2 Loss of Generative Preconditions and the Definitional Crisis of Structural Disappearance

In traditional mathematical systems, every mathematical symbol implicitly carries specific structural preconditions for its generation. However, once these symbols are incorporated into axiomatic systems, the demands of formalization and systematic expansion often lead to excessive abstraction and axiomatization, which can override and detach them from their original generative structures.

This process ultimately places mathematical definitions into a structural vacuum in which their underlying connections are no longer preserved. The result is a **definitional crisis caused by structural disappearance**: mathematical objects remain formally defined within the axiomatic system, while the generative structures and conditions that originally gave rise to those definitions have been removed from consideration.

2. FUNDAMENTAL PRINCIPLES OF STRUCTURE OVER AXIOMS

2.1 The Hierarchical Relationship Between Structure and Axioms

In traditional mathematics, axioms are regarded as unprovable and absolute starting points. Under the framework of **Structure Over Axioms**, however, structure is not merely an abstract system of symbolic logic, but an objectively existing **Universal Matrix**.

Structure as the Matrix:
Each dimension possesses an intrinsic and non-negotiable structural matrix of its own. For example, the two-dimensional basis is centered on the natural logarithmic base e, which supports arithmetic operations and self-similarity, while the three-dimensional basis is governed by a rotationally expanding spiral structure.

Axioms as Projections:
Axioms are localized rules extracted by humans to describe the underlying matrix through a lower-dimensional cross-section. Structure determines the boundaries within which axioms remain valid; the structural authority of the matrix therefore takes precedence over any human-defined axiomatic system.

2.2 The Nature of Dimension: Basis and Expansion

The evolution of structure follows a strict hierarchy of dimensions. Its fundamental mechanism can be understood as the dynamic coupling between **Basis** and **Expansion**.

The Hard Constraint of the Basis:
The basis is not an arbitrary coordinate system. It is the underlying structural code through which the matrix expresses self-similarity and generates fundamental operators, such as $+, -, \times, \div$.

Multidimensional Projection of Expansion:
As the basis expands outward, it gives rise to higher-dimensional geometries and phase manifolds. For example, the two-dimensional matrix structure can, through specific phase rotations and expansions, develop upward into the three-dimensional spiral matrix. This bottom-up generative relationship is governed entirely by the intrinsic constraints of numbers and does not permit arbitrary modification by axiomatic rules.

2.3 Strict Structural Constraint: Dimensional Degeneration

The highest criterion for determining whether a structure constitutes a genuine underlying matrix lies in its **Dimensional Degeneration**.

When the additional degrees of freedom of a higher-dimensional structure approach zero—that is, when the dimensional parameter is reduced toward 0 — the structure must not leave behind a logical discontinuity or divergent residue. Instead, it must degenerate losslessly and precisely into the complete lower-dimensional matrix.

For example, when the three-dimensional spiral structure undergoes degeneration, it must converge coherently and correspond precisely to the complete two-dimensional e-based structural matrix. This cross-dimensional **degenerative self-consistency** replaces the conventional reliance on human-defined patches for maintaining consistency and becomes one of the strongest structural constraints of the proposed mathematical framework.

2.4 Fundamental Criteria for Structure Over Axioms

Once the principles of the **Universal Matrix** and **Dimensional Degeneration** have been established, the evaluation of a mathematical system no longer depends primarily on the accumulation of axioms, but returns to three fundamental structural criteria:

Matrix Self-Similarity:
Whether operators and rules can be naturally derived from the Universal Matrix of the given dimension—such as e or the spiral structure—rather than being externally imposed.

Degenerative Losslessness:
Whether, when a higher-dimensional system degenerates toward a lower dimension, its structural boundaries and matrix definitions remain fully continuous and free from truncation error.

Phase–Basis Compatibility:
Whether the entire computational process remains constrained by the three-dimensional basis and its phase structure, thereby eliminating structural distortions arising from discrete positional numeral systems.

3. $\tan(90^\circ)$: THE DEFINITIONAL PROBLEM WHEN THE PREREQUISITE STRUCTURE DISAPPEARS

3.1 The Conventional Preconditions for $\tan(90^\circ)$

Within the conventional theory of trigonometric functions, the tangent function is defined as the ratio

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}. \tag{1}$$

As θ approaches 90°, this definition depends on algebraic division over the real numbers together with the corresponding limiting operation. Since

$$\cos(90^\circ) = 0, \tag{2}$$

the denominator vanishes and the conventional algebraic framework can no longer assign an ordinary real value to the expression. Accordingly, $\tan(90^\circ)$ is conventionally classified as **undefined**, while the limiting behavior of $\tan(\theta)$ as $\theta \to 90^\circ$ is described as divergence toward **infinity**.

This distinction already reveals an important limitation of the conventional framework: the symbolic rule is preserved, but the underlying geometric conditions that generate the ratio are no longer explicitly represented within the resulting expression.

3.2 Angle, Direction, and Projection Structure

From a structure-prior perspective, an **angle** is not merely an isolated scalar value, but a **phase-projection state** within a rotational matrix.

The value 90° is therefore not simply an isolated numerical point. It represents a limiting state of orthogonal projection in which the active vector becomes perpendicular to the reference projection plane, causing the corresponding component on that cross-section to vanish.

At this point, the numerical value is better understood not merely as a computational result, but as a **manifestation node** necessarily generated by the rotational manifold at a specific phase cross-section.

3.3 Can $\tan(90^\circ)$ Remain Defined When Its Prerequisite Structure Disappears?

When the conventional expression loses its denominator because

$$\cos(90^\circ) = 0, \tag{3}$$

the conventional axiomatic framework cannot directly represent the corresponding geometric boundary through ordinary division. It therefore treats the point as **undefined**, while the surrounding limiting behavior is characterized by divergence toward ∞.

From the perspective of the **Universal Matrix** and phase transformation, however, the issue is not necessarily the creation of a physical entity of infinite magnitude. Rather, the coordinate cross-section reaches a boundary characterized by **phase degeneracy and dimensional transition**.

The apparent divergence of $\tan(90^\circ)$ can therefore be interpreted as a forced truncation produced by a lower-dimensional projection tool—such as Cartesian coordinates and ratio-based division—when confronted with an orthogonal boundary.

If the underlying continuous rotational matrix is restored, the point itself remains part of the manifold. What disappears is not necessarily the underlying structural state, but the ability of the conventional axiomatic representation to describe that state.

3.4 The Inseparability of Value, Symbol, and Structure

The problem raised by $\tan(90^\circ)$ exposes a fundamental blind spot of conventional mathematics: **the separation of symbols and numerical values from their generative structures**.

When $\tan(90^\circ)$ is treated as an isolated symbol that can simply be substituted into an algebraic expression, the notions of undefinedness, divergence, and infinity emerge as separate formal consequences. Once **Structure Over Axioms** is established as the governing principle, however, numerical values and symbols cannot be separated from their generative preconditions and phase boundaries.

Under this framework, many phenomena conventionally described as **singularities** or **divergences toward infinity** may not represent a failure of the underlying reality itself. Rather, they may indicate the failure of a lower-dimensional axiomatic representation at a structural boundary.

4. INFINITY, BOUNDARIES, AND STRUCTURAL EXTENSION

4.1 Infinity as a Structural Boundary

Traditional mathematics has long treated **Infinity** as a mysterious ultimate quantity, a dynamically unending progression, or a philosophical problem that has generated extensive debate among mathematicians, such as the distinction between actual infinity and potential infinity.

Within the framework of **Structure Over Axioms**, however, infinity is not fundamentally regarded as an objectively existing entity. Rather, it functions as a **system warning signal**: an indication that the current representational tool has reached its operational boundary.

When a computation, a truncation imposed by a positional numeral system, or a lower-dimensional projection exceeds the effective domain of an axiomatic system, the system returns signals such as divergence or undefinedness. From this perspective, this response constitutes the structural meaning of infinity.

4.2 Finite Structures and Infinite Extension

This warning mechanism provides a different perspective on the longstanding question of whether mathematics is **discovered** or **invented**.

Discovery:
Within the region in which the infinity warning has not been triggered, our computations and derivations of mathematical regularities are fundamentally acts of discovery. They reveal structural relationships that already exist within the given **Universal Matrix** and its lower-dimensional projections.

Invention:
Once the infinity boundary is reached, the existing tools have declared their limitations. Mathematical progress can therefore no longer consist merely of treating divergence as a failure. Instead, it must **embrace the boundary and invent new tools**.

The act of crossing such a boundary and constructing new higher-dimensional operators or dimensional frameworks constitutes the genuinely inventive aspect of mathematical development.

4.3 Necessary Conditions for Structural Extension

When an existing mathematical tool triggers an infinity warning, structural extension becomes a necessary requirement for maintaining logical self-consistency. However, extension cannot be an unconstrained act of arbitrary construction.

Its primary condition is the **inheritance of the structural constraints of the preceding matrix**. Any newly constructed tool or higher-dimensional framework must strictly satisfy the principle of **Dimensional Degeneration**, ensuring that when the additional degrees of freedom approach zero, the new structure can degenerate losslessly and precisely into the complete lower-dimensional matrix.

No artificial patch or logical discontinuity may be introduced during this process.

4.4 Determinacy of Structural Extension

When confronting an unknown domain beyond an established boundary, how can the direction of extension be determined without falling into arbitrary or random possibilities?

The answer lies in **Evolutionary Convergence**.

Structural evolution is not an unrestricted divergence of possibilities. It is strongly guided by the underlying **Universal Matrix** and its symmetries. When multiple possible paths advance toward higher dimensions, only those paths that satisfy **self-similarity** and **phase continuity** can remain structurally compatible with one another.

This internally generated convergence of structural evolution provides the extension process with determinacy and establishes its scientific necessity.

4.5 Arbitrary Extension vs. Structural Extension

Based on the principles above, two fundamentally different paths of mathematical development can be distinguished:

Arbitrary Extension:

The uncontrolled accumulation of formal axioms, such as disconnected generalizations commonly found in highly abstract mathematical systems. Without the constraints of an underlying matrix or the guidance of evolutionary convergence, such extension ultimately leads toward increasingly complex logical constructions accompanied by structural distortion.

Structural Extension:

An extension rooted in the **Universal Matrix**, constrained by **Dimensional Degeneration**, and guided by **Evolutionary Convergence**. When the infinity warning is triggered, new mathematical tools are constructed precisely at the structural boundary while preserving the constraints inherited from the preceding dimensional level.

This constitutes the legitimate path by which the proposed mathematical framework can continue to develop and, ultimately, provide a means of structurally deconstructing physical reality.

5. MATHEMATICAL CRISES UNDER STRUCTURE OVER AXIOMS

5.1 The Common Structural Problem Underlying Mathematical Crises

A review of the three major crises in the history of mathematics shows that conventional accounts have often attributed them to gaps in formal logic or deficiencies in definitions, followed by attempts to repair the mathematical system through additional axioms or stricter definitions.

From the metatheoretical perspective of **Structure Over Axioms**, however, these three crises can be interpreted as manifestations of a common structural problem:

> **The incompatibility and boundary activation that inevitably arise when low-dimensional, discrete axiomatic tools attempt to represent higher-dimensional, continuous matrix structures.**

From this perspective, each major adjustment made by conventional mathematics in response to a foundational crisis can be understood as a defensive modification of axiomatic systems in response to structural distortion.

5.2 The First Mathematical Crisis: Irrational Numbers and the Conflict with Continuity

Limitations of the Conventional Interpretation:

The Pythagorean school attempted to characterize geometric magnitudes through ratios of rational numbers. When the incommensurability of $\sqrt{2}$ was revealed, Greek mathematics was forced to distinguish geometric reasoning from numerical representation, eventually leading to the expansion of the number system to accommodate such quantities.

The Essence from a Structure-Prior Perspective:

The rational-number system fundamentally functions as a discrete positional and ratio-based projection tool, whereas geometric continuity is an intrinsic property of the underlying manifold.

The emergence of $\sqrt{2}$ is therefore not a defect of the numerical value itself, but a **structural discontinuity produced when a discrete positional grid attempts to map a continuous matrix**.

From this perspective, defining the resulting quantity as an **irrational number** merely assigns a symbolic label to the boundary. It does not address the underlying difference between the phase structures of the numerical representation and the geometric matrix.

5.3 The Second Mathematical Crisis: The Foundations of Calculus and Infinitesimals

Limitations of the Conventional Interpretation:

Early calculus relied heavily on the manipulation of **infinitesimals** (dx). Their algebraic treatment created a logical tension in which an infinitesimal appeared to behave as both zero and nonzero. This issue was eventually addressed through the rigorous reconstruction of calculus using the ϵ–δ framework.

The Essence from a Structure-Prior Perspective:

From the present framework, an infinitesimal is not merely an imprecise transitional symbol. Rather, it can be interpreted as a **transient differential cross-section of a higher-dimensional phase manifold during dynamic evolution**.

The algebraic difficulties encountered by early calculus arose, from this perspective, because a dynamic phase manifold was forced into a static framework of discrete ratios.

The structure-prior interpretation instead regards calculus as the natural differential evolution of a matrix structure across dimensional expansion. Its self-consistency therefore originates from the structure itself rather than being imposed afterward through a formal limit framework.

5.4 The Third Mathematical Crisis: Formal Systems and Incompleteness

Limitations of the Conventional Interpretation:
From Russell's paradox to Gödel's incompleteness theorems, foundational developments challenged the expectation that formal systems could establish their own completeness and consistency in the absolute sense envisioned by Hilbert's program. They thereby revealed intrinsic limitations of formalized deductive systems.

The Essence from a Structure-Prior Perspective:
Within the structure-prior framework, Gödel's incompleteness results can be interpreted as having a clear topological implication:

> **A low-dimensional axiomatic formal system cannot fully contain, within itself, the higher-dimensional matrix to which it belongs.**

Axioms are fundamentally local, low-dimensional projections, whereas the underlying matrix is treated as a holistic and potentially unbounded structure.

From this perspective, the result demonstrates the limitations of relying exclusively on axiomatic deduction. The proposed framework instead seeks overall structural self-consistency through a return to the **Universal Matrix** and structural evolution.

5.5 Structural Comparison of the Three Mathematical Crises

Mathematical Crisis	Conventional Mathematical Response	Interpretation Under Structure Over Axioms
First (Irrational Numbers)	Expansion of the number system and introduction of the concept of irrational numbers to fill the apparent gap between numerical representation and geometric magnitude.	**Structural boundary activation** produced when a discrete positional grid attempts to map a continuous manifold.
Second (Calculus)	Abandonment of infinitesimals as foundational objects and development of the rigorous ϵ–δ theory of limits.	**Transient cross-sections and operator-projection evolution** arising during the dynamic evolution of a phase manifold across dimensions.
Third (Incompleteness)	Recognition of the incompleteness of formal systems and abandonment of the goal of establishing absolute consistency through a single formal framework.	The **fundamental limitation of low-dimensional axiomatic projections in representing their higher-dimensional matrix**.

5.6 The Structural Mechanism and Common Origin of Mathematical Crises

Taken together, the three crises suggest a fundamental blind spot in the conventional development of mathematics:

> **Low-dimensional regional projections are mistaken for the complete underlying matrix, and the resulting limitations are addressed through iterative additions of formal axioms intended to preserve systemic closure.**

As such patch-based corrections accumulate, mathematical systems may progressively move further away from the underlying physical reality they are intended to describe.

The **Structure Over Axioms** framework developed in this paper therefore rejects this model of post hoc repair. Instead, it takes the **Universal Matrix** and **Dimensional Degeneration** as foundational constraints, providing a mathematical construction path intended to connect continuously with the underlying structure of physical reality.

6. PMM AS A CONCRETE CASE OF MATHEMATICAL STRUCTURE

6.1 From Structural Principles to PMM

The preceding sections established the metatheoretical principle of **Structure Over Axioms** and the evolutionary logic of the **Universal Matrix**. However, any mathematical innovation must ultimately be grounded in concrete physical reality and a corresponding computational framework.

In the derivation of **Space Interlayer Conjecture / Phase Manifestation Mechanics (SIC / PMM)**, spacetime is not treated as a passive flat container or manifold in the conventional geometric sense. Instead, it is regarded as a dynamic holographic structure possessing intrinsic phase characteristics and a unique temporal sampling base frequency.

This section therefore uses PMM as a concrete case to demonstrate how a new mathematical structure can be generated directly from underlying structural principles rather than assembled through post hoc axiomatic construction.

6.2 Mathematical Requirements of the PMM-Specific Equations

To precisely describe the evolution of spacetime possessing phase characteristics and three-dimensional spiral geometry, conventional calculus and purely scalar differential equations become insufficient.

The mathematical formulation of PMM must directly incorporate **phase (θ)**, **frequency (ω)**, and the core coupling parameters, expressing macroscopic time $\mathcal{S}_T$ as the computational result of the underlying **Universal Matrix** $\mathbb{U}$ under specific constraints.

Its minimal foundational equation can be expressed as

$$\mathcal{S}_T = [\mathbb{U}(\beta = 0.0865)]_{\omega}^{\theta}\,. \tag{4}$$

Here, $\beta = 0.0865$ serves as a key structural constant governing the coupling strength of the system at the relevant scale. The frequency ω in the subscript and the phase θ in the superscript jointly define the boundary conditions of the multidimensional matrix operation, thereby moving beyond the conventional algebraic treatment in which dependent and independent variables are treated as separate entities.

6.3 Limitations of Existing Mathematical Representations

When describing spacetime evolution, existing mathematical frameworks encounter fundamental representational limitations:

- **Passivity of the conventional treatment of time:**
  Conventional relativity and quantum mechanics treat time primarily as a continuous coordinate parameter t, while overlooking the possibility that spacetime itself may possess a structural constraint associated with a discrete sampling base frequency, represented here by the **central-axis step** Λ.

- **Distortion through geometric projection:**
  When an overall three-dimensional spiral-like spacetime manifold is forcibly reduced and embedded into Cartesian coordinates or a flat spacetime metric, higher-dimensional phase characteristics may be truncated and consequently appear as various unnecessary singularities and divergent terms.

This suggests that without fundamentally reconstructing the underlying mathematical structure, deeper regularities of physical reality may remain obscured by the limitations of existing symbolic systems.

6.4 Conditions for Constructing the New Mathematical Structure

For PMM and its subsequent four-dimensional extension to be established, the construction of the new mathematical structure must satisfy the following core conditions:

1. **Phase Dominance:**
   Each computational unit must directly carry phase and frequency variables, so that a numerical value is no longer treated as an isolated scalar but as a phase matrix possessing holographic directional information.

2. **Compatibility with the Central-Axis Base Frequency:**
   Numerical evolution must correspond to the underlying sampling base frequency, represented by the central-axis step Λ. This ensures that the evolution of macroscopic time $\mathcal{S}_T$ follows strict structural stepping constraints rather than an infinitely divisible smooth abstraction.

3. **Degenerative Self-Consistency:**
   When the mathematical structure is projected toward lower dimensions or approaches a parameter limit, it must losslessly recover the preceding matrix structures, such as the two-dimensional e structure and the three-dimensional spiral basis. This ensures seamless self-consistency throughout the theoretical framework.

6.5 Algorithmic Unlocking: The Fine-Grained Downward Derivation of Λ

Within the overall framework of the three-dimensional forward spiral, the central-axis base frequency Λ is not an arbitrarily assigned abstract parameter. Rather, it is the unique orthogonal solution of the system under multidimensional dynamic projection.

Its underlying structure is generated through the coupling between the strongly constrained lattice 11.56 (i.e., 3.4^2) and the minimum non-resonant path α, together with the holographic projection of the exponential self-similarity of the three-dimensional basis e.

The core generative equation of this fine-grained downward derivation can be expressed as

$$\Lambda = 11.56 \times \alpha \times e^{3.4\alpha}. \tag{5}$$

Through this algorithm, Λ integrates the strongly constrained lattice, the non-resonant path, and exponential self-similarity into a unified structure. It demonstrates how the new mathematical framework can directly use matrix operators and crystalline geometric constraints to lock the sampling step of spacetime.

This construction thereby moves beyond the conventional dependence on manually fitted numerical approximations or discrete assumptions, providing the evolution of macroscopic spacetime with an internally generated algorithmic foundation.

7. BASE OMNIDIMENSIONAL PROJECTION GEOMETRY

7.1 The Original Extension Rule and Regional Constraints

During the initial construction of the mathematical formulation of PMM, we introduced a fixed coupling constant, such as $\beta = 0.0865$, as a reference anchor in order to constrain the physical reality and crystalline structure at a specific scale. Although this **hard assignment** successfully constrained the correspondence between the central-axis stepping and the phase matrix during the initial stage, it implicitly restricted the dynamic spacetime manifold to a specific static cross-section, thereby limiting the possibility of a full structural expansion of the system.

7.2 Modification of the Key Extension Condition: Replacing the Constant with Λ

To overcome the limitations of a local special case and move toward a broader global manifold evolution, the extension condition must be fundamentally modified: **isolated constants and unconstrained generalized variables are both discarded, and the sampling base frequency of the central-axis step, Λ, is instead directly adopted as the core independent variable of the Universal Matrix.**

Through this modification, macroscopic time $\mathcal{S}_T$ no longer depends on the arbitrary substitution of a numerical value. Instead, it is directly driven by the base frequency Λ generated from the underlying crystalline structure and the fine-grained downward derivation algorithm. Its core generative equation is formally evolved into:

$$\mathcal{S}_T = [\mathbb{U}(\Lambda)]^{\theta}_{\omega} \tag{6}$$

In this formulation, Λ, as an intrinsically constrained independent variable of the matrix, carries the holographic information of the strongly constrained lattice and exponential self-similarity, thereby replacing the causally ungrounded scalar variables of conventional mathematics.

7.3 Structural Changes After the Modification

When the internal kernel of the equation is transformed from β into a dynamic structure governed by Λ, the mathematical system undergoes a fundamental transition:

- **Unification of Crystalline Structure and Equation:**
  The flow of macroscopic time ($\mathcal{S}_T$) and the microscopic crystalline stepping ($\Lambda = 11.56 \times \alpha \times e^{3.4\times\alpha}$) are seamlessly coupled at the operator level, providing the mathematical derivation with rigorous support from physical reality.
- **Precise Convergence of Topological Degrees of Freedom:**
  The system no longer requires artificial patches to correct scale discrepancies, because Λ itself constitutes the unique orthogonal solution of the system under multidimensional dynamic projection, ensuring absolute continuity in structural expansion.

7.4 Stability and Consistency of the New Structure

When the independent variable is upgraded to the base frequency Λ, conventional mathematics may raise the concern that introducing additional uncontrolled variables could cause the system to diverge. However, under the framework of **Structure Over Axioms**, this new structure exhibits a high degree of stability:

- The value of Λ is not arbitrarily assigned, but is strongly constrained by the minimum non-resonant path α and the self-similarity of the underlying matrix.
- As the system is extended toward boundary limits, the evolution of Λ must still strictly obey **Dimensional Degeneration**, ensuring lossless recovery of the lower-dimensional mother structure.

This design—**with Λ as the core driver and structural constraints as the boundary of admissible evolution**—eliminates the logical distortions of conventional mathematics when dealing with complex spacetime manifolds and establishes the foundation for subsequent extensions toward higher-dimensional architectures.

8. FORMATION OF THE THREE-DIMENSIONAL BASIS MATRIX AND THE DISCRETE SAMPLING MATRIX

8.1 Generative Conditions of the Three-Dimensional Basis and Discrete Sampling

In conventional geometry, three-dimensional space is often presupposed as a pre-existing and passive continuous Cartesian coordinate container. Under the framework of **Structure Over Axioms**, however, the three-dimensional basis is by no means an arbitrarily given background. Rather, it is an inevitable product of the full-dimensional expansion of the Universal Matrix under specific phase constraints.

Its fundamental characteristic lies in the establishment of a **Discrete Sampling Point Set (Sampling Point Set,** $\mathcal{K}_\Lambda$). The generation of the three-dimensional basis must introduce crystalline constraints governed by the central-axis stepping base frequency Λ and the minimum non-resonant path α. Space is therefore no longer regarded as an infinitely divisible continuous manifold, but as a node matrix composed of intrinsic rotational degrees of freedom and sampling characteristics.

8.2 Three-Dimensional Spiral Structure and Matrix Properties

Why must the three-dimensional basis necessarily exhibit a **spiral-like** structure rather than a simple linear extension or a closed circle?

A purely linear extension would cause the system to diverge without bound, whereas a purely closed rotation would become trapped in an infinite loop. The three-dimensional spiral structure simultaneously accommodates dynamic progression and phase self-consistency: it uses the central-axis stepping Λ as its sampling base frequency while performing circumferential rotation through phase θ and frequency ω.

This matrix property gives space an intrinsic rotational momentum and a discrete sampling mechanism, allowing the three-dimensional basis to possess a computationally tractable and dynamically evolvable structural capacity from the outset.

8.3 Dimensional Degeneration and the Lower-Dimensional Mother Structure

As a critical link between dimensional levels, the three-dimensional basis matrix must strictly obey the core principle of **Dimensional Degeneration**.

When the additional rotational degrees of freedom or pitch parameters of the three-dimensional spiral structure approach their limiting values, or when certain parameters approach zero, the structure must not produce computational divergence or a logical discontinuity. Instead, it must **losslessly and precisely degenerate into a lower-dimensional mother structure**, such as the two-dimensional e-based matrix.

This downward compatibility through degeneration demonstrates that the three-dimensional basis is not an isolated special case, but a stable node within a continuous bottom-up evolutionary chain.

8.4 Coordinates and Discrete Representation of the Three-Dimensional Basis

Conventional coordinate systems divide space into mutually disconnected continuous scalar scales, severing the structural relationships between individual points. In contrast, the mathematical representation of the three-dimensional basis must return to the underlying nature of structure.

Within this framework, any node in the three-dimensional basis is no longer an isolated static numerical value, but a manifestation of the Universal Matrix $\mathbb{U}(\Lambda)$ on a specific sampling lattice $\mathcal{K}_\Lambda$.

Its coordinate system is directly constructed from the base frequency Λ, phase θ, and frequency ω, such that the spatial coordinates themselves carry complete information about dynamic evolution and crystalline constraints.

8.5 Treatment Beyond the Sampling Boundary and Holography

Within the architecture of the discrete sampling point set, the treatment of spatial boundaries becomes highly direct: **regions outside the sampling points constitute a purely invalid information region (Null Domain) and are directly ignored by the system**, thereby fundamentally eliminating the problematic infinite divergences encountered in conventional field theories.

At the same time, the three-dimensional basis matrix possesses a decisive property of **Holography**. Because the matrix is generated through exponential self-similarity and a strongly constrained lattice, any local sampling cross-section within the three-dimensional basis in fact completely maps the structural logic of the entire matrix. This ensures absolute smoothness and seamless coupling when the framework is subsequently extended toward higher dimensions.

9. THE UNIQUE NATURE OF THE OPERATOR AND THE DYNAMIC Λ THRESHOLD

9.1 The Nature of the Operator: The Unique Interference Operation ($\circledast$)

Traditional mathematics relies on various externally defined operators, such as addition, subtraction, multiplication, division, differentiation, integration, and matrix multiplication, to address different classes of problems. This approach is analogous to decomposing a complex function through a Taylor expansion into a collection of numerous elementary algebraic terms.

However, within the metatheory of **Structure Over Axioms**, the universal generative substrate continuously evolves through a three-dimensional spiral governed by the fundamental frequency Λ. Accordingly, the only primitive operation at the deepest level of the system is the **interference operation ($\circledast$)**.

Here, the term "unique" does not deny the validity of differentiation, integration, matrix operations, or other operations in conventional mathematics. Rather, it indicates that these operations are not mutually independent as ultimate primitives. Instead, they are derived expressions generated by interference relations within the generative substrate under different phase, frequency, and structural conditions.

Thus, all apparently distinct mathematical transformations, dimensional projections, and structural interactions may fundamentally be regarded as recursive expansions of the same underlying operator, $\circledast$, under different conditions. The dynamic interweaving of structures can be expressed as:

$$\mathcal{S}_{T1} \circledast \mathcal{S}_{T2} \tag{7}$$

Here, $\circledast$ is the underlying engine of self-interference within the generative substrate. It does not depend upon an externally imposed toolbox of independent operations; rather, it emerges from the phase relations, sampling conditions, and geometric constraints intrinsic to the structure itself.

9.2 Phase-Aware Overlap Range (Overlap Range, $\mathcal{O}$)

When the interference operation

$$\mathcal{S}_{T1} \circledast \mathcal{S}_{T2} \tag{8}$$

is performed, it is first necessary to determine whether an effective intersection boundary exists between the two structural states. This boundary is referred to as the **Overlap Range**.

The Overlap Range is not merely a static coordinate intersection. It is a composite intersection region jointly determined by the **discrete sampling point sets** and the **local phase-superposition conditions**. In other words, the two structures form an effective overlap only when they possess a geometric intersection at the sampling points and their local phases satisfy the conditions for a valid structural interweaving.

Let the discrete sampling point sets of the two structural states be denoted by $\mathcal{K}_{\Lambda 1}$ and $\mathcal{K}_{\Lambda 2}$, respectively. Their effective overlap range can then be formalized as:

$$\mathcal{O}(\mathcal{S}_{T1}, \mathcal{S}_{T2}) \equiv \{p \in \mathcal{K}_{\Lambda 1} \cap \mathcal{K}_{\Lambda 2} \mid \theta_1(p) \oplus \theta_2(p) \text{ satisfies the conditions for valid local phase interweaving}\} \tag{9}$$

Here, $\oplus$ denotes the phase-level combination formed by the interference of the two local phases. It is defined as:

$$\theta_1 \oplus \theta_2 \equiv \mathcal{P}_\theta \left(\mathcal{S}_{T1} \circledast \mathcal{S}_{T2}\right) \tag{10}$$

where $\mathcal{P}_\theta$ is a derived projection operator acting on the phase degree of freedom. It projects the underlying interference result into the phase degree of freedom, producing a **Net Phase Superposition** state.

Therefore, $\oplus$ is not an additional independent primitive operator. It is a derived representation of $\circledast$ at the phase level.

The existence condition for an effective overlap is:

$$\mathcal{O}(\mathcal{S}_{T1}, \mathcal{S}_{T2}) \neq \emptyset \tag{11}$$

This indicates that at least one effective sampling node exists at which both the geometric intersection condition and the phase-interweaving condition are simultaneously satisfied.

Within the present framework, space and phase are not separate externally imposed descriptions. They are intrinsically coupled at discrete sampling points. Any region that does not correspond to an effective sampling point, or that fails to satisfy the required phase correspondence, is assigned to the **Null Domain** and filtered out in subsequent operations.

9.3 The Unlocking Threshold for Structural Coupling: Resonance Range and the Λ Ratio

Once the effective overlap range

$$\mathcal{O} \neq \emptyset \tag{12}$$

has been established, the system introduces the dimensionless Λ ratio, χ_Λ, as a dynamic threshold for determining whether the interacting structures can achieve resonance locking:

$$\chi_\Lambda = \frac{\Delta\Psi}{\Lambda} = \frac{\theta_1 \oplus \theta_2}{\Lambda} \tag{13}$$

where $\Delta\Psi$ denotes the phase-combination result generated by the local interference of the two structural states.

Non-Resonant Region

When

$$\chi_\Lambda \in \mathbb{R} \setminus \{k\alpha\} \tag{14}$$

the interference result manifests as cancellation, phase slippage, or ineffective interweaving, and the structures cannot establish strong coupling.

Resonance-Locking Region

When

$$\chi_\Lambda = k\alpha, \qquad k \in \mathbb{Z} \tag{15}$$

the system enters a **resonance-locked state**, triggering strong orthogonal coupling and enabling constructive interweaving and higher-order structural evolution.

Accordingly, structural interaction may be represented sequentially as:

$$\text{Overlap} \longrightarrow \text{Resonance} \longrightarrow \text{Coupling} \tag{16}$$

The Overlap Range determines whether the two structures possess the geometric conditions required for interaction, while the Λ ratio further determines whether that interaction can enter a resonance-locked state.

9.4 Geometrization of Information Flow: Recursive Expansion of $\circledast$ and Topological Manifold Deformation

Within the effective overlap range $\mathcal{O}$, the interaction between structures is not a static numerical superposition. Instead, it is a dynamic structural reconfiguration driven by $\circledast$.

Through recursive expansion of $\circledast$, the local phase combination

$$\theta_1 \oplus \theta_2 \tag{17}$$

drives the flow and transmission of information to manifest directly as geometric deformation of the topological manifold.

Essential Transformation

Traditional mathematics generally represents information as an abstract scalar, vector, or probability distribution carried over a spatial domain. Within the present framework, however, information flow is identified with phase reconfiguration occurring at discrete sampling points.

This reconfiguration alters the local structure, curvature, and geometric boundaries of the three-dimensional spiral generative substrate. Information is therefore not an independent content superimposed upon geometry; it is the dynamic variation of the geometric structure itself.

In this sense:

Geometry is structure; information is deformation.

Mathematical Formulation

Through the accumulation of interference at discrete sampling points, the structural deformation of the manifold, $\Delta\mathcal{M}$, can be expressed as:

$$\Delta\mathcal{M} = \sum_{p\in\mathcal{O}} \left(\mathcal{S}_{T1} \circledast \mathcal{S}_{T2}\right)_p \tag{18}$$

This expression indicates that the local interference results within the effective overlap range accumulate discretely to produce the overall structural deformation of the manifold.

Accordingly, matter, information, and spatial geometry are no longer treated as three separate objects. Rather, they are manifestations of the same underlying structure at different levels of description.

9.5 The Minimum Non-Resonant Path α and the Integer-Multiple Discrete Resonance Grid (Preliminary Framework for the Prime-Number Problem)

Within the resonance criterion, the unlocking condition derived from the **minimum non-resonant path** α carries a potentially important correspondence to number-theoretic structure.

Generation of the Resonance Grid

The parameter α, acting as an underlying constraint of the generative substrate, restricts the lockable conditions in phase space to a discrete set of integer-multiple resonance points:

$$\chi_\Lambda = k\alpha, \qquad k \in \mathbb{Z} \tag{19}$$

These points form an **Integer-Multiple Discrete Resonance Grid**.

The grid is not generated merely by uniform partitioning of a continuous domain. Instead, it is jointly determined by the exponential self-similarity of the generative substrate, the discrete sampling conditions, and the structural constraints.

Strategic Significance

In subsequent number-theoretic research, prime numbers may therefore be examined not merely as abstract irreducible integers, but as **structural damping points or phase barriers that fail to form integer-multiple resonance locking within the current Λ grid**.

This statement does not presuppose that prime numbers have already been completely decomposed within the framework. Rather, it proposes a structural hypothesis that may be formally investigated in subsequent work: the irreducibility found in number theory may correspond to a form of structural non-decomposability or resonance-unlocking condition within the geometric resonance system.

Accordingly, this section establishes the underlying conceptual framework for future investigations of prime distributions and related number-theoretic problems through geometric resonance, discrete resonance grids, and structural irreducibility.

10. FORMAL REPRESENTATION OF THREE-DIMENSIONAL BASE MATHEMATICS

10.1 Basic Symbols and Definitions

Before proceeding to the formal mathematical derivation of the three-dimensional base, the core symbols and endogenous definitions of the present theoretical framework are first formalized and consolidated. A fundamental distinction must be made here between **the free parameters of the mathematical base** and **the specific manifestation conditions of the physical world**.

- **The Universal Substrate** $\mathbb{U}(\Lambda)$: the generative substrate carrying holographic information and exponential self-similarity. Here, Λ represents the axial stepping base frequency adopted by the substrate. Therefore, $\mathbb{U}(\Lambda)$ is not a single fixed object, but a family of structures corresponding to different Λ bases.
- **The Axial Stepping Base Frequency** Λ: a free base parameter in three-dimensional base mathematics, used to specify the discrete sampling scale, rotational stepping, and structural resolution. Within pure mathematical formalization, Λ is not assumed to be a fixed constant; its value may vary according to the selected representational basis, resolution scale, or structural conditions.

  When three-dimensional base mathematics is further mapped onto the physical world, Λ is no longer an arbitrary mathematical degree of freedom, but may instead be selected by the constraints of the physical structure. For example, under the physical conditions corresponding to the present framework, one may obtain:

$$\Lambda_{\text{phys}} = 11.56 \times \alpha \times e^{3.4\times\alpha} \tag{20}$$

  Accordingly, this value represents a **specific manifestation condition of the physical world**, rather than an initial axiom or fixed constant of three-dimensional base mathematics itself.
- **The Minimum Non-Resonant Path** α: the underlying constraint parameter of the crystalline structure, which determines the fundamental spacing and resonance conditions of the discrete resonance grid.
- **The Structural Spacetime State** $\mathcal{S}_T$: the dynamic manifestation of the universal substrate under specific phase θ, frequency ω, and base parameter Λ.
- **The Unique Interference Operation** $\circledast$: the unique primitive operator at the underlying level of the system, used to describe interweaving, interference, and resonance between structural states.
- **The Phase-Aware Overlap Range** $\mathcal{O}$: the effective region jointly determined by the intersection of discrete sampling points and the conditions for local phase interweaving.
- **The** Λ **Ratio** χ_Λ: the dimensionless dynamic resonance threshold formed by the local phase combination relative to the current base parameter Λ:

$$\chi_\Lambda = \frac{\Delta\Psi}{\Lambda} \tag{21}$$

Thus, at the mathematical level, Λ is first treated as a **freely selectable structural base**. At the physical level, it may subsequently be manifested as a specific value through the constraints of the substrate structure and the physical world. These are different levels of description and should not be conflated.

10.2 Mathematical Encoding of the Three-Dimensional Base

Within the structure-first framework, the three-dimensional base is not a single fixed continuous Cartesian coordinate container. Instead, it is a discrete sampling structure defined by the selected base parameter Λ.

Three-dimensional base mathematics therefore first describes a **family of structures parameterized by the free base parameter** Λ, rather than presupposing a unique physical scale.

1. Discrete Sampling Point Set ($\mathcal{K}_\Lambda$)

For any given Λ, the three-dimensional base is composed of discrete nodes generated by that base parameter and its rotational trajectories:

$$\mathcal{K}_\Lambda = \{p_i \mid p_i = \mathbb{U}(\Lambda)[\theta_i, \omega_i], \quad i \in \mathbb{N}\} \tag{22}$$

Here, Λ determines the axial stepping and sampling scale of the current mathematical representation.
When Λ changes, the corresponding discrete sampling structure may change accordingly. Thus:

$$\Lambda_1 \neq \Lambda_2 \quad \Longrightarrow \quad \mathcal{K}_{\Lambda_1} \neq \mathcal{K}_{\Lambda_2} \tag{23}$$

In general, different values of Λ do not represent different underlying principles, but different discrete representations of the same structure under different base scales.

2. Three-Dimensional Spiral Coordinate Correspondence

Any sampling node p_i is not merely a scalar position, but a structural state carrying complete information about its dynamic evolution:

$$\mathcal{S}_T(p_i) = [\mathbb{U}(\Lambda)]^{\theta}_{\omega} \tag{24}$$

Therefore, spatial coordinates are no longer merely passive positional markers, but simultaneously possess the capacity for structural description, computation, and evolution.
The significance of this representation does not lie in fixing a particular value of Λ, but in establishing a three-dimensional mathematical representation that can vary with the base parameter while preserving the underlying structural relations.

The following figures illustrate three geometric modes within a shared axial structural framework, consisting of a central axial spine, discrete Λ-indexed sampling nodes, radial ribs connecting the nodes to the central axis, and spiral trajectories extending along the axis:

(a) Constant-Radius Spiral:

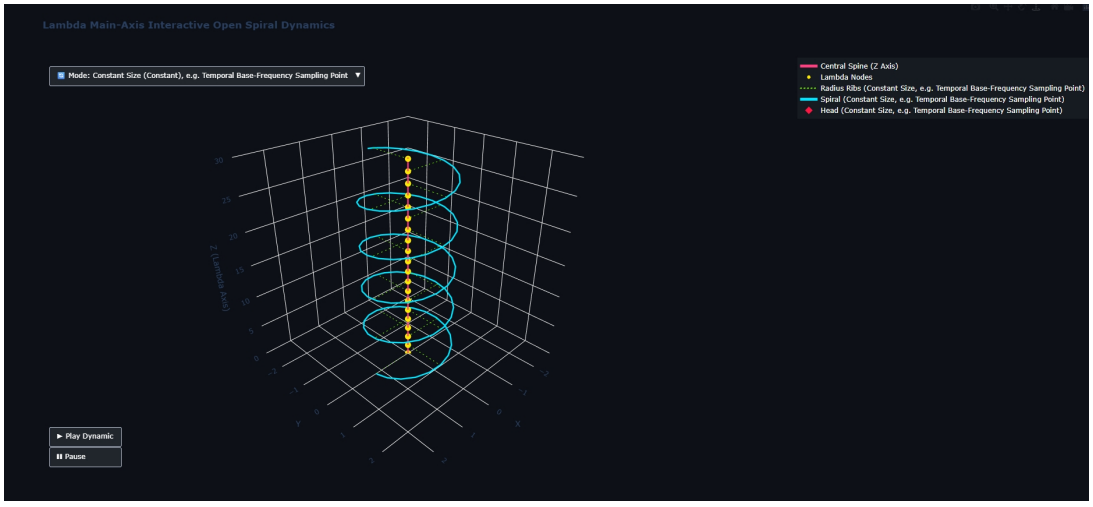


FIG. 1. A geometric illustration of a constant radius, with Λ serving as a temporal base-frequency sampling point.

(b) Contracting Spiral:

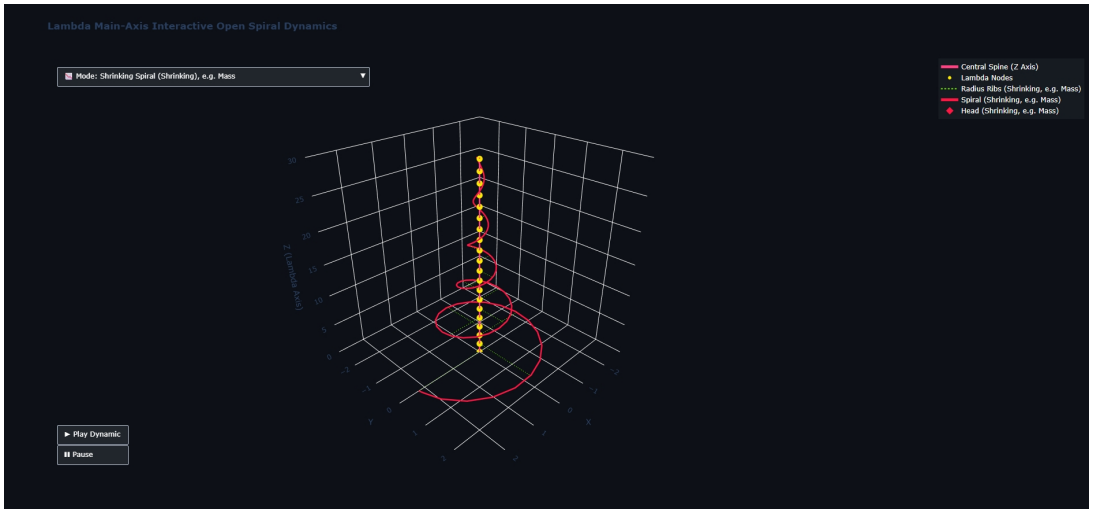


FIG. 2. The radius decreases as the axial parameter increases, illustrating the conceptual correspondence in Phase Manifestation Mechanics (PMM) between mass and closed rotation.

(c) Open Expanding Spiral:

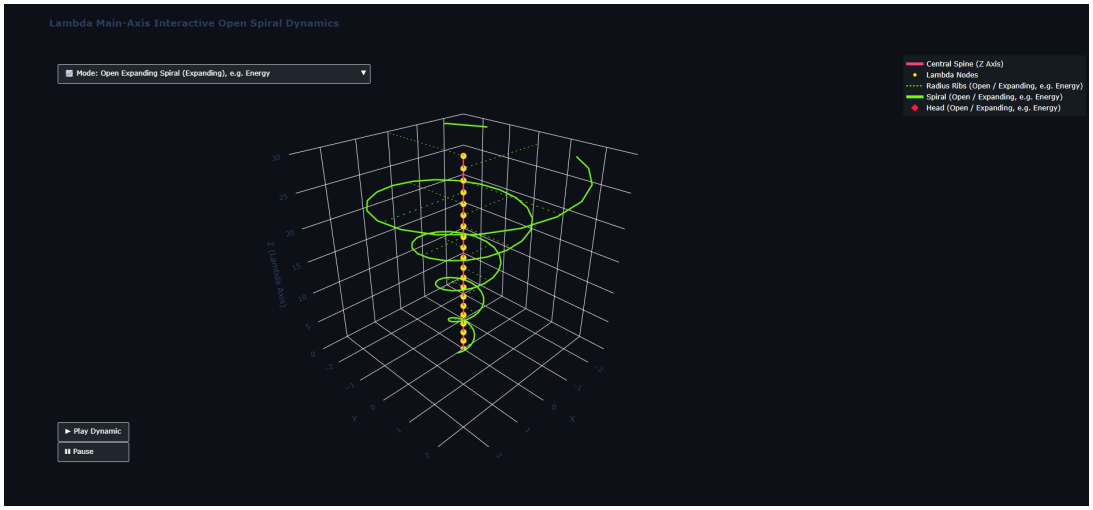


FIG. 3. The radius increases as the axial parameter increases, illustrating the conceptual correspondence in PMM between energy and open rotation.

These three modes are not independent structures, but distinct forms of radial development within a common axial geometric framework.

The value $\Lambda = \pi/2$ is selected solely to make the discrete sampling intervals visually distinguishable and does not represent the physical parameter $\Lambda_{\text{phys}} = 0.0865$. All radial functions, coefficients, and coordinate ranges are illustrative choices for geometric visualization. They do not represent calibrated physical quantities, experimentally measured values, or quantitative predictions derived from BOPG or PMM.

The purpose of these figures is to illustrate structural relationships and geometric configurations rather than to constitute a quantitative physical simulation. The interactive version allows users to select and animate the three geometric modes independently.

10.3 Representation of Structural Relations and Operations

Interactions between structures are no longer described through mutually independent primitive operators, but are unified through $\circledast$ together with its boundary conditions.

Since Λ is a free base parameter at the mathematical level, all resonance conditions involving Λ should first be understood as **structural relations relative to the currently selected base**.

1. Overlap Test

The first step is to determine whether two structures possess an effective overlap:

$$\mathcal{O}(\mathcal{S}_{T1}, \mathcal{S}_{T2}) \neq \emptyset \tag{25}$$

If no effective overlap exists, the corresponding region is assigned to the Null Domain and filtered out in subsequent operations.

2. Resonance Range and Λ-Ratio Criterion

Within the effective overlap range, calculate:

$$\chi_\Lambda = \frac{\theta_1 \oplus \theta_2}{\Lambda} \tag{26}$$

If

$$\chi_\Lambda = k\alpha, \qquad k \in \mathbb{Z} \tag{27}$$

constructive locking is triggered; otherwise, the interaction manifests as cancellation, phase slippage, or structural degradation.

Here, Λ is the base parameter selected for the current mathematical representation. Therefore, the resonance condition describes:

the locking relation of the structure relative to a specific base.

Only when the model is further mapped onto the physical world does the free parameter Λ need to be specialized into a specific Λ_{phys} determined by physical structural conditions.

10.4 Derivation Process Under Structural Constraints

The standardized derivation process of the present framework can be represented as follows:

1. **Substrate and Base Setting**: specify the underlying constraint parameter α and select the base parameter Λ for the current three-dimensional mathematical representation.
2. **State Generation**: generate the structural spacetime state $\mathcal{S}_T$ and its corresponding discrete sampling point set $\mathcal{K}_\Lambda$ through the universal substrate.
3. **Interference and Interweaving**: perform the unique primitive interference operation between structural states:

$$\mathcal{S}_{T1} \circledast \mathcal{S}_{T2} \tag{28}$$

4. **Boundary Filtering**: identify the effective region $\mathcal{O}$ satisfying both geometric intersection and local phase-interweaving conditions, and filter out the Null Domain.

5. **Resonance Determination and Manifold Deformation**: use χ_Λ to distinguish the resulting evolutionary paths and, through recursive expansion of the interference operation, calculate the topological-geometric deformation $\Delta\mathcal{M}$ induced by information flow.

6. **Physical Mapping (Optional)**: when the three-dimensional base mathematics is applied to the physical world, specialize Λ according to the corresponding physical structural constraints, thereby obtaining a specific Λ_{phys}.

Accordingly, the relationship between mathematical derivation and physical manifestation can be explicitly distinguished as:

$$\text{Free Base} \longrightarrow \text{Structural Derivation} \longrightarrow \text{Physical Constraint} \longrightarrow \text{Specific Manifested Value} \tag{29}$$

10.5 The Fundamental Difference of Structure Over Axioms

This formal representation proposes a foundational generative perspective that differs from conventional mathematics:

- **Structure Precedes Externally Imposed Axioms**: the present framework does not take externally imposed axioms as the ultimate generative source of all rules. Instead, operational rules are generated from the internal structure and constraints of the universal substrate.

- **Base Parameters Precede Specific Values**: within three-dimensional base mathematics, Λ is first treated as a free structural parameter. A specific numerical value is not supplied a priori, but is selected through additional structural and physical constraints when the mathematical structure is mapped onto the physical world.

- **Discrete Sampling Precedes Continuous Extension**: the framework takes the discrete sampling point set $\mathcal{K}_\Lambda$, generated under the selected Λ base, as the fundamental descriptive unit rather than presupposing that space must first possess an infinitely divisible continuous structure.

- **Unification of the Primitive Operator**: differentiation, integration, matrix operations, and related procedures are no longer regarded as mutually independent primitive sources, but are understood as derived expressions of the unique interference operation $\circledast$ under different structural conditions.

Therefore, the core of three-dimensional base mathematics does not lie in prescribing a fixed set of constants in advance or in introducing additional externally imposed rules. Instead, it lies in establishing a **formal system in which the base parameter may vary freely, while the operational relations can be generated intrinsically from the structure of the universal substrate**.

On this basis, when the mathematical structure is mapped onto the physical world, a specific Λ appears as a manifestation of the corresponding physical structure rather than as a prior constraint imposed upon three-dimensional base mathematics.

11. BOPG: STRUCTURAL REPRESENTATION, Λ-GRID MAPPING, AND CONTINUOUS RECONSTRUCTION

BOPG proposes a structural representation centered on the mother structure, sampling basis, phase relations, and grid mappings. Its primary purpose is to examine:

> Whether the same structure may present different mathematical forms under different representation systems, sampling bases, and resolutions.

From this perspective, different representation systems may be regarded as different levels or paths for describing a structure. The correspondence, transformation, inclusion, or limitation between these representations must be determined through specific structures and mappings.

11.1 Structure and Representation

The fundamental starting point of BOPG is:

$$\text{Structure} \neq \text{A Single Representation of the Structure} \tag{30}$$

A given structure may be represented through different coordinates, scales, phases, grids, or computational operations. When the conditions of representation change, properties such as continuity, discreteness, finiteness, infiniteness, convergence, divergence, or singularity may also need to be reconsidered under the new representational conditions.

The central questions examined by BOPG are:

1. What are the fundamental conditions on which a representation depends?
2. Which structural relations can that representation preserve?
3. Which phenomena originate from the structure itself?
4. Which phenomena originate from the sampling method or the range of representation?
5. Whether a traceable structural mapping exists between different representations.

Accordingly, mathematical problems can be examined through the following relation:

$$\text{Mother Structure} \longrightarrow \text{Sampling Basis} \longrightarrow \text{Structural Representation} \longrightarrow \text{Phase Relations} \longrightarrow \text{Grid Mapping} \tag{31}$$

11.2 Structural Mapping and Continuous Reconstruction Between Λ-Grids

In BOPG, the grid $\mathcal{K}_\Lambda$ is not a fixed and invariant coordinate background. Rather, it is a structural representation jointly formed by the mother structure $\mathbb{U}$, the sampling basis Λ, the phase state θ, and the rotational state ω:

$$\mathcal{K}_\Lambda = \mathcal{K}(\mathbb{U}, \Lambda, \omega, \theta) \tag{32}$$

Different values of Λ do not merely represent scale variations on the same grid. They may also correspond to different:

- Sampling resolutions;
- Degrees-of-freedom configurations;
- Node distributions;
- Phase representations;
- Structural representation methods.

When Λ changes, the effective sampling grid of the system may accordingly be adjusted, reconstructed, or reconfigured. This change may be regarded as a re-manifestation of the mother structure under a different sampling basis.

For any two sampling bases, Λ_1 and Λ_2, BOPG examines whether a structural correspondence exists between them:

$$\mathfrak{T}_{\Lambda_1 \to \Lambda_2} : \mathcal{K}_{\Lambda_1} \longrightarrow \mathcal{K}_{\Lambda_2} \tag{33}$$

The mapping $\mathfrak{T}_{\Lambda_1 \to \Lambda_2}$ need not be a linear scaling. It may also involve:

1. Reconfiguration of sampling nodes;
2. Re-representation of phase structures;
3. Adjustment of local resolution;
4. Redistribution of degrees of freedom;
5. Preservation or transformation of structural relations;
6. Information correspondence between different grids.

Within BOPG, "continuity" may be understood in a generalized sense:

> When the sampling basis Λ changes, can the structure maintain its internal relations, phase order, and traceability through a series of mutually corresponding grid states?

If a parameterized family of grids exists,

$$\left\{\mathcal{K}_{\Lambda(s)}\right\}_{s \in I}, \tag{34}$$

then continuous reconstruction may be represented as:

$$\mathcal{K}_{\Lambda(s_1)} \xrightarrow{\mathfrak{T}} \mathcal{K}_{\Lambda(s_2)} \xrightarrow{\mathfrak{T}} \mathcal{K}_{\Lambda(s_3)} \tag{35}$$

Its central concern is whether mappings exist between different grids that are definable, traceable, and capable of preserving key structural relations.

Accordingly, BOPG approaches finiteness, infiniteness, divergence, and representational boundaries by examining whether these properties may be re-expressed as the Λ-grid changes.

11.3 The Scope of Structural Representation and Conditions of Transformation

Within BOPG, the following three levels must be distinguished:

1. The Structure Itself

Whether a given phenomenon or relation exists within the mother structure under investigation.

2. A Specific Representation

How that structure is expressed under a particular set of coordinates, scales, grids, phases, or computational rules.

3. The Mapping Between Representations

Whether different representations can be transformed into one another, and which structural relations can be preserved during the transformation.

Therefore:

$$\text{Inexpressible in a Given Representation} \neq \text{Nonexistence of the Structure} \tag{36}$$

Likewise:

$$\text{Representable in BOPG} \neq \text{Proven Equivalent to Another Representation} \tag{37}$$

For any given problem, BOPG must separately examine:

- How the problem is defined within its original representation;
- Whether a corresponding structure exists within BOPG;
- Whether an explicit mapping exists between the two representations;
- Whether the mapping preserves the necessary structural relations;
- Whether the transformed result retains the same mathematical or physical meaning.

The relationship between different mathematical representations may take the form of:

$$\text{Correspondence} \quad , \quad \text{Transformation} \quad , \quad \text{Inclusion} \quad , \quad \text{Restriction} \quad \text{or} \quad \text{Not Yet Established} \tag{38}$$

These relationships should be determined through subsequent formalization and specific mappings.

11.4 Structural Examination of Major Mathematical Problems Through BOPG

BOPG examines selected major mathematical problems within the framework of structural representation, sampling bases, and grid mappings.

Here, "correspondence" means:

> Whether the problem can be given a new representation within BOPG that possesses structural significance.

Problem	**Direction of Examination in BOPG**	**Status of Relation**
Poincaré Conjecture	Three-dimensional manifolds, topological closure, and structural contraction	Correspondence to be established
Navier–Stokes Problem	Fluid motion, local reorganization, and sampling resolution	Model to be established
Riemann Hypothesis	Prime distribution, resonance grids, and structural arrangement	Correspondence to be established
P vs NP	Search spaces, parallel structures, and global locking	Formal relation to be established

Poincaré Conjecture

BOPG examines whether a structural correspondence exists between three-dimensional manifold problems and the closure, contraction, phase consistency, and topological representation of three-dimensional structures.

Key questions include:

- How can BOPG structural states represent three-dimensional manifolds?
- Does structural contraction correspond to a certain topological operation?
- Is there a relationship between phase consistency and topological invariants?
- Are closure conditions equivalent under different representations?

Navier–Stokes Problem

BOPG treats fluid motion as the dynamic reorganization of structure under different sampling scales, local phases, and coupling conditions.

Under this representation, singular behavior in the equations may correspond to:

- Limitations of the current sampling resolution;
- Local structures exceeding the established range of representation;
- The need to reconfigure the grid;
- The system entering another phase or coupling state.

These are candidate structural interpretations. Whether a formal relation can be established with the Navier–Stokes equations and their regularity problem must be determined through specific mappings and proofs.

Riemann Hypothesis

BOPG examines whether a relationship exists between prime distribution and resonance grids, phase arrangements, and structural damping.

One candidate direction is:

> Certain numerical structures may be unable to decompose into lower-order stable structures because of their phase or resonance conditions.

Whether this description has a formal equivalence with the irreducibility of prime numbers requires further definitions, mappings, and proofs.

P vs NP

BOPG examines the structural differences involved in computational complexity through search spaces, parallel paths, local verification, and global structural locking.

The issues to be addressed include:

- How search spaces are generated by structures;
- Whether structures possess decompositions that can be parallelized;
- Whether local verification can guarantee a global solution;
- Whether grid mappings can change the effective search range;
- How BOPG structural states correspond to standard computational models.

11.5 α as a Structural Dynamic Parameter

Within BOPG, α is a dynamic parameter jointly formed by the mother structure $\mathbb{U}$, the sampling basis Λ, the phase state, and the resonance conditions:

$$\alpha = \alpha(\mathbb{U}, \Lambda, \omega, \theta) \tag{39}$$

The significance of α lies not only in its numerical value, but also in its role and constraints within the overall structure. Its central question is:

$$\text{Is } \alpha \text{ jointly determined by the mother structure, the sampling grid, and the resonance conditions?} \tag{40}$$

If α exhibits a numerical correspondence with a physical constant, that correspondence may serve as an entry point for structural investigation. Further work is nevertheless required to examine:

- The origin of the correspondence;
- Units and dimensionless conditions;
- Numerical error;
- Parameter uniqueness;
- Whether it can be necessarily derived from the mother structure and Λ.

11.6 Renormalization, Analytic Continuation, and Cross-Grid Reconstruction

Within BOPG, scale transformations and continuation operations may be reconsidered as structural mappings and continuous reconstructions between different Λ-grids.

Existing Concept	**Direction of Examination in BOPG**
Renormalization	Scale and structural mapping between different Λ-grids
Analytic continuation	Continuation of phase structures beyond the original representational range
Divergence	Structural overflow under specific grid or representational conditions
Continuity	Structural traceability between different discrete sampling states
Scale transformation	Re-representation of structure after changing the sampling basis

Renormalization

BOPG examines renormalization in the form:

$$\mathcal{K}_{\Lambda_1} \longrightarrow \mathcal{K}_{\Lambda_2} \tag{41}$$

Its possible contents include the redistribution of degrees of freedom, changes in phase scales, resampling of local structures, re-representation of effective parameters, and the preservation or transformation of structural relations between different grids.

Analytic Continuation

BOPG examines analytic continuation as follows:

> When a representation fails within a specific range, can the original phase relations be extended through structural mapping to another representable grid or structural region?

The main considerations include:

- Whether the original structural relations are preserved;
- Whether the continued representation is compatible with the original grid;
- Whether a traceable phase correspondence exists between different grids;
- Whether the continuation leads to a new structural state;
- Whether the continuation process involves the Null Domain.

Divergence and the Null Domain

Within BOPG, divergence may be examined as structural overflow under a particular sampling grid or representational condition. This phenomenon may correspond to:

$$\text{Failure of the Current Representation} \quad \text{or} \quad \text{The Need to Reconfigure the Grid} \tag{42}$$

The Null Domain serves as a conceptual tool for describing regions that cannot be effectively represented under the current sampling conditions. Its specific definition, boundary conditions, grid-reconstruction mechanism, and relationship to convergence require further formalization.

BOPG therefore proposes the following direction of examination:

$$\boxed{\text{Structural Mapping and Continuous Reconstruction Between Different } \Lambda\text{-Grids}} \tag{43}$$

11.7 Directions for Further Formalization

To allow BOPG's structural representation to be further examined and tested, the following elements must be established:

1. A formal definition of the primitive interference operator $\circledast$;
2. The generation method, effective range, and reconstruction conditions of $\mathcal{K}_\Lambda$;
3. The mapping between different Λ-grids:
$$\mathfrak{T}_{\Lambda_1 \to \Lambda_2} \tag{44}$$
4. Conditions under which the mapping preserves overlap, resonance, coupling, and necessary information;
5. The convergence, reversibility, or equivalence conditions of grid reconstruction;
6. The formal relationship between $\Lambda(s)$ and the discrete sampling set $\mathcal{K}_{\Lambda(s)}$;
7. The correspondence between BOPG structural states and differentiation, integration, topology, and computational models;
8. Independent formalization pathways for each major mathematical problem.

The central purpose of this section is to establish a structural interface capable of accommodating different representation systems:

$$\boxed{\text{Structure} \longrightarrow \text{Sampling Basis} \longrightarrow \Lambda\text{-Grid} \longrightarrow \text{Structural Mapping} \longrightarrow \text{Continuous Reconstruction}} \tag{45}$$

The primary research direction of BOPG is:

> Whether phenomena that appear as finite, infinite, divergent, singular, or non-extendable under a specific representation can obtain another consistent representation through structural mapping and continuous reconstruction between different Λ-grids.

This direction does not predetermine the conclusion. Its focus remains on the structure, the representations, and the mappings themselves.

12. THE GEOMETRIC NECESSITY OF PROBABILITY: FROM DETERMINISTIC MATRICES TO SECTIONAL DISTRIBUTIONS

12.1 Structural Limitations of the Conventional Concept of Probability

Conventional probability theory and statistical mechanics are built upon explicit axiomatic and measure-theoretic structures, including the **Kolmogorov axioms**, σ**-algebras**, and the abstract **sample space** Ω. Whether through the limit theorems employed in frequentist statistics or the subjective belief representation adopted in Bayesian statistics, their common feature is the use of probability measures to describe outcomes that remain undetermined under a given set of observational conditions.

From the perspective of **Structure Over Axioms**, the issue is not whether these probabilistic tools are valid, but rather whether **probability must be regarded as a fundamental source of randomness at the deepest level of nature**.

- **Fundamental BOPG Perspective**: within Base Omnidimensional Projection Geometry (BOPG), the Universal Substrate $\mathbb{U}(\Lambda)$ and the unique interference operator $\circledast$ are regarded as deterministic underlying structures. Within this framework, "randomness" is not treated as a fundamental entity independent of structure, but rather as **a low-dimensional statistical manifestation produced when an observer encounters a higher-dimensional structure that has not yet been fully resolved**.

Therefore, an observed probabilistic event does not necessarily imply that the underlying structure itself possesses irreducible randomness. Instead, it may arise from the statistical manifestation of higher-dimensional phase relations, structural coupling, and projection conditions within a finite observational section.

In other words:

> **It is not that no describable structure exists; rather, the observer can access only a low-dimensional section of that structure.**

12.2 Higher-Dimensional Geometry and Lower-Dimensional Sections

Why do macroscopic phenomena and microscopic measurements exhibit various probability distributions? Within BOPG, this is not primarily interpreted as the result of nature performing random sampling, but as:

> **the geometric volume-density distribution produced when a higher-dimensional structural manifold is projected onto a lower-dimensional observational section.**

1. Volume Projection Effect

When a higher-dimensional structural spacetime state $\mathcal{S}_T$ is projected through the discrete sampling point set $\mathcal{K}_\Lambda$ into a lower-dimensional observational space, different curvatures, phase densities, and local overlap degrees within the higher-dimensional structure can produce non-uniform geometric accumulation on the lower-dimensional section.

Therefore, the same higher-dimensional structure may produce different sectional distributions under different observational sections or different sampling bases.

2. Conversion of Geometric Density into Probability Density

Within the BOPG description, the **Probability Density Function (PDF)** defined in conventional statistics can be reinterpreted as **the normalized sectional representation of the geometric density generated by the projection of the higher-dimensional structure within the effective overlap range**.

If $\rho_{\mathrm{geom}}(x)$ denotes the geometric density projected onto the observational section, then the corresponding probability density can be written as:

$$P(x) = \frac{\rho_{\mathrm{geom}}(x)}{\displaystyle\int \rho_{\mathrm{geom}}(x)\, dx} \tag{46}$$

Under this representation, probability is no longer treated as an abstract quantity independent of structure, but as a statistical quantity generated by geometric density through sectional projection and normalization.

Regions with higher distribution peaks correspond to higher effective overlap or projection density of the higher-dimensional structure, whereas the tails and boundary regions correspond to lower-density regions, stronger curvature variation, or regions approaching the geometric boundary within the structural projection.

12.3 Phase Projection and the Geometric Generation of the Normal Distribution

During the interference operation

$$\mathcal{S}_{T1} \circledast \mathcal{S}_{T2} \tag{47}$$

if the local phase superposition $\theta_1 \oplus \theta_2$ fails to lock precisely onto the resonance grid defined by $k \cdot \alpha$, the system enters a non-resonant path characterized by phase slippage and dissipation.

When multiple geometrically incommensurate, three-dimensional helical phase perturbations are superposed and projected onto a lower-dimensional observational section, their statistical distribution can be described through the mathematical limit of deterministic phase projection. Under appropriate centering and scale normalization, the global interference of these uncoupled helical trajectories naturally converges to a normal form.

1. Accumulation of Multiple Incommensurate Helical Perturbations

Let the net phase deviation on a lower-dimensional observational section be composed of N uncoupled, geometrically incommensurate microscopic discrete sampling interference terms ξ_i generated by the underlying helical structures:

$$S_N = \sum_{i=1}^{N} \xi_i \tag{48}$$

where each ξ_i is structurally constrained by the grid base frequency Λ and local phase boundaries. Crucially, within this deterministic framework, the expectation operator $\mathbb{E}[\cdot]$ is defined not as a stochastic ensemble average, but as a geometric spatial average over the continuous helical trajectories in phase space (reflecting phase ergodicity). Assuming each term possesses a finite spatial mean and variance, we define:

$$\mathbb{E}[\xi_i] = \mu_i, \qquad \mathrm{Var}(\xi_i) = \sigma_i^2 \tag{49}$$

Owing to the geometric incommensurability and mutual uncoupling of the helical phases, the cross-correlation terms vanish under global sectional integration. Thus, the total spatial mean and variance of the sum are respectively given by:

$$\mu_N = \sum_{i=1}^{N} \mu_i, \qquad \sigma_N^2 = \sum_{i=1}^{N} \sigma_i^2 \tag{50}$$

To study the limiting geometric distribution on the observational section, we define the centered and scale-normalized phase variable:

$$X_N = \frac{S_N - \mu_N}{\sigma_N} \tag{51}$$

2. Characteristic Function and Destructive Suppression of Higher-Order Geometric Ripples

By virtue of phase-ergodic independence among the incommensurate helical components, the characteristic function of X_N can be expressed as a product of the individual phase characteristic functions. Its logarithmic form is therefore given by the corresponding sum of logarithmic characteristic functions:

$$\ln \phi_{X_N}(t) = \sum_{i=1}^{N} \ln \mathbb{E}\left[e^{\,it(\xi_i-\mu_i)/\sigma_N}\right] \tag{52}$$

Under the structural constraints of BOPG, the minimum non-resonant path α restricts the effective combinations of phase perturbations, ensuring that individual variances σ_i^2 remain bounded. As $N \to \infty$, the total variance $\sigma_N \to \infty$, driving the higher-order Taylor expansion arguments to zero.

Geometrically, this implies that the micro-ripples and high-frequency structural irregularities contributed by individual helical phases undergo mutual destructive interference upon projection. Consequently, the higher-order cumulant terms become progressively suppressed relative to the first- and second-order terms, and the dominant contribution on the observational section approaches a smooth quadratic form:

$$\ln \phi_{X_N}(t) \longrightarrow -\frac{1}{2}t^2 \tag{53}$$

More generally, if we map this back to the unscaled coordinate system while retaining the non-centered mean term, the limiting form of the phase-space characteristic function can be written as:

$$\ln \phi_X(t) = i\mu t - \frac{1}{2}\sigma^2 t^2 \tag{54}$$

whose corresponding characteristic function is Gaussian.

The mathematical limit described here has a formal correspondence with the normal convergence described by the **Central Limit Theorem** in conventional probability theory. Within BOPG, however, this convergence is stripped of any stochastic interpretation; it is understood purely as a geometric limit produced when a higher-dimensional, fully deterministic helical structure undergoes extensive local phase projection and lower-dimensional sectionalization.

It should be emphasized that the above construction is intended to display the minimal conditions under which a normal form appears on the lower-dimensional section, rather than to exhaust the higher-dimensional definition of the structure. Two conditions are essential: the helical components must be mutually incommensurate in the global sense, that is, no nontrivial integer linear relation exists among their frequencies; and the resonance range of the structure must remain finite, so that the superposed helices are not locked into a single coherent configuration.

3. Geometric Manifestation of the Normal Distribution

Applying the Fourier inverse transform to the Gaussian characteristic function yields the normal probability density on the lower-dimensional observational section:

$$P(X) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(X-\mu)^2}{2\sigma^2}\right) \tag{55}$$

Therefore, within the BOPG framework, the normal distribution is not regarded as being generated merely by external random errors, but can instead be reinterpreted as:

> **the stable statistical-geometric limit formed when a higher-dimensional deterministic structure is projected onto a lower-dimensional observational section through a large number of discrete phase perturbations.**

In this sense, the normal convergence described by conventional probability theory may, within BOPG, be regarded as a sectional distribution manifested by the same underlying deterministic structure under projection across different dimensional descriptions.

12.4 Geometric Conditions for Probability Generation and Fat-Tail Truncation

Within the BOPG framework, any event expressed in probabilistic form can be further traced to its corresponding geometric overlap conditions and structural boundaries.

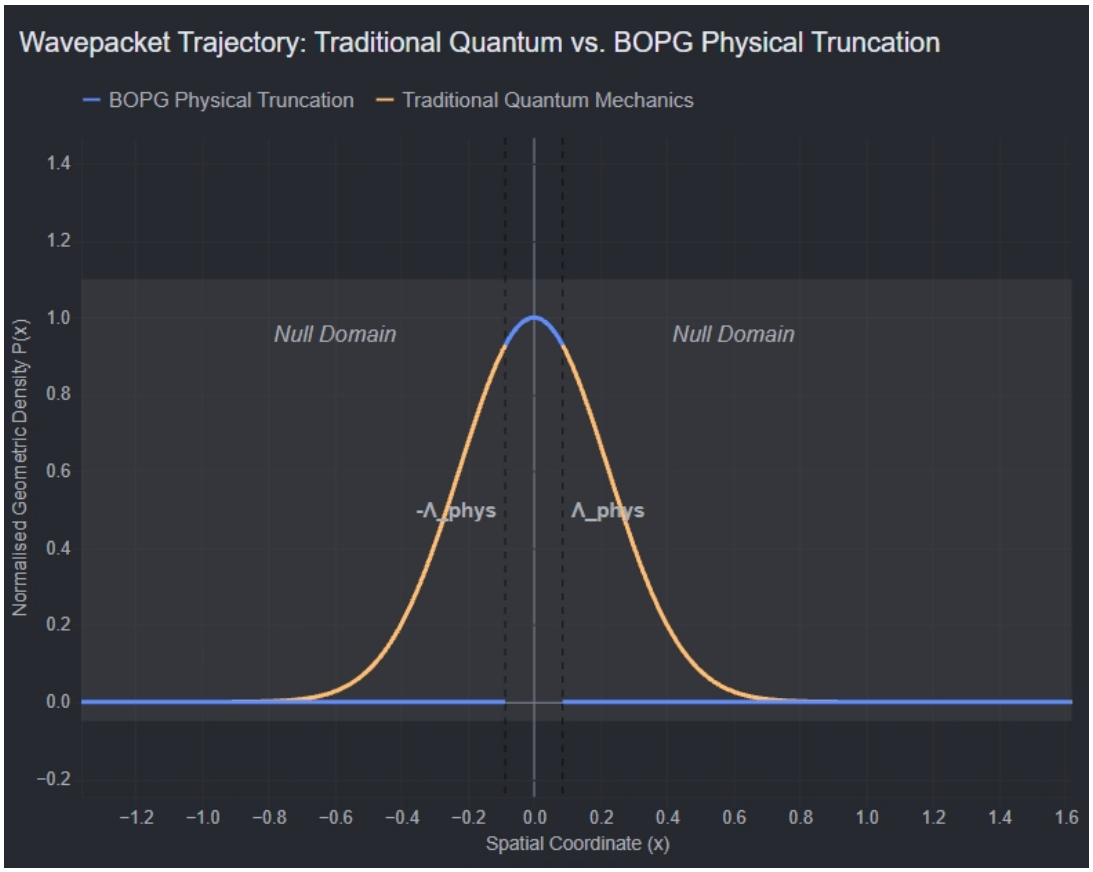


Figure 12.4: Geometric Illustration of Physical Manifestation Truncation in BOPG

This figure compares the representational difference between the complete Gaussian distribution and its truncation within the BOPG physical manifestation domain. The orange curve represents the complete Gaussian distribution, retaining its original mathematical form. The blue curve represents the distribution constrained by the physical manifestation domain, preserving the geometric density corresponding to the original Gaussian distribution within the manifestation domain, while no physical manifestation occurs outside this domain, which is classified as the Null Domain.

Within the BOPG framework, this distinction does not modify or eliminate the Gaussian distribution. Instead, it distinguishes the complete mathematical distribution from the finite region capable of physical manifestation. The structural relationship can be expressed as:

$$\mathcal{D}_{\text{manifest}} = \mathcal{D}_{\Lambda_{\text{phys}}} \cap \mathcal{R}_{\text{phys}}^{\text{exp}} \tag{56}$$

Here, $\Lambda_{\text{phys}} = 0.0865$ represents the basis scale associated with physical manifestation in reality, while $\mathcal{R}_{\text{phys}}^{\text{exp}}$ denotes the physical resonance range to be determined experimentally. Together, these two conditions define the physical manifestation domain; regions outside this domain are classified as the Null Domain in this representation.

The vertical dashed lines and the $\pm\Lambda_{\text{phys}}$ markers indicate the truncation boundaries schematically. They do not represent an experimentally established resonance width, nor do they imply that all actual resonance boundaries can be predetermined solely from Λ_{phys}.

This figure is intended to illustrate the structural distinction between a mathematical distribution and its physical manifestation range. It does not modify the Gaussian distribution itself, nor does it claim to establish quantitative experimental validation.

1. Resonance Volume Ratio

Within the effective overlap range $\mathcal{O}$, let $\mathcal{O}_{\mathrm{res}}$ denote the effective region satisfying the resonance condition

$$\chi_\Lambda = k\alpha, \qquad k \in \mathbb{Z} \tag{57}$$

and let $\mathcal{O}_{\mathrm{total}}$ denote the total effective overlap region. The probability corresponding to the event can then be expressed as a ratio of geometric volumes:

$$P(\mathrm{Event}) = \frac{\mathrm{Vol}\left(\mathcal{O}_{\mathrm{res}}\right)}{\mathrm{Vol}\left(\mathcal{O}_{\mathrm{total}}\right)} \tag{58}$$

In this sense, probability is no longer merely an abstract measure value, but may be interpreted as **the ratio of the geometric volume satisfying a specific structural locking condition to the total effective volume under given observational conditions and within the effective structural domain**.

Thus, a probability value can be regarded as the geometric proportion of a structural condition within the observational section.

2. Physical Λ and Fat-Tail Truncation

In pure mathematical representation, Λ is a free base parameter. Different values of Λ may therefore correspond to different sampling scales and effective resolution domains. Whether a fat tail is subject to truncation depends on the selected base and its corresponding structural boundaries.

When BOPG is further mapped onto the physical world, Λ no longer remains arbitrarily free, but is specialized to a specific value by the actual conditions of physical manifestation:

$$\Lambda_{\mathrm{phys}} = 0.0865 \tag{59}$$

This physical-world base frequency provides a concrete discrete stepping scale and structural resolution boundary, thereby imposing a corresponding effective sampling domain on physically manifestable states.

Let

$$\mathcal{K}_{\mathrm{phys}} = \mathcal{K}_{\Lambda_{\mathrm{phys}}} \tag{60}$$

denote the effective physical sampling domain generated under $\Lambda_{\mathrm{phys}} = 0.0865$. If the tail of a probability distribution extends beyond the effective manifestation range permitted by this structure, it can no longer be resolved within the same base and is instead assigned to the Null Domain.

Thus, under the physical mapping, one obtains:

$$\mathrm{supp}(P_{\mathrm{phys}}) \subseteq \mathcal{K}^{\mathrm{eff}}_{\mathrm{phys}} \tag{61}$$

That is, **the unlimited extension of a fat tail is not necessarily an inevitable feature of physical structure, but may be subject to truncation by the discrete geometric scale and effective structural boundary associated with $\Lambda_{\mathrm{phys}} = 0.0865$.**

The key point is not to claim that $\Lambda = 0.0865$ automatically eliminates all fat tails in abstract mathematics. Rather, the point is that once Λ is specialized to 0.0865 by the structural conditions of the physical world, the base that was mathematically free to extend acquires a concrete physical resolution scale and boundary.

Accordingly, the central BOPG perspective on fat-tail phenomena is:

> **An infinite mathematical tail may exist within the formal representation of a free base; however, when the structure is mapped onto the physical world with a specific Λ_{phys}, the physical manifestation domain becomes subject to the corresponding geometric scale limitation.**

This truncation mechanism may provide a basis for further investigation of extreme events, non-Gaussian distributions, financial risk, and high-energy tails in physical systems.

13. FROM SECTIONAL DISTRIBUTIONS TO QUANTUM GEOMETRIC TRAJECTORIES: SCHRÖDINGER MANIFESTATION UNDER Λ TRUNCATION

13.1 From Theoretical Resonance Conditions to Physical Resonance Domains

Chapter 9 established the fundamental resonance condition between structures:

$$\chi_\Lambda = \frac{\Delta\Psi}{\Lambda} \tag{62}$$

When

$$\chi_\Lambda = k\alpha, \qquad k \in \mathbb{Z} \tag{63}$$

the system enters a theoretical resonance-locking condition.

Chapter 10 further defined Λ as a free parameter of three-dimensional base mathematics and distinguished the mathematical base from the physical manifestation conditions. When the structure is mapped onto the physical world, Λ may be specialized by physical conditions to:

$$\Lambda_{\text{phys}} = 0.0865 \tag{64}$$

However, the theoretical resonance condition initially describes the **resonance position and locking relation of the structure**, and does not directly determine the actual width of a resonance region in a physical system.

Real systems possess finite resolution, coupling strength, damping, and boundary conditions. Therefore, resonance in the physical world may appear as an effective region with finite width rather than as a single mathematical point.

Accordingly, the physical resonance condition may be further expressed as:

$$|\chi_\Lambda - k\alpha| \le \delta_{\text{res}} \tag{65}$$

where δ_{res} represents the allowable resonance deviation of the physical system.

At the purely mathematical level, δ_{res} does not need to be fixed in advance. When the model is mapped onto a physical system, it may instead be treated as a physical quantity that can be measured experimentally.

Thus, an experimentally determined physical resonance domain may be defined as:

$$\mathcal{R}^{\text{exp}}_{\text{phys}} = \left\{ x \;\middle|\; \left|\chi_{\Lambda_{\text{phys}}}(x) - k\alpha\right| \le \delta^{\text{exp}}_{\text{res}} \right\} \tag{66}$$

Here, $\mathcal{R}^{\text{exp}}_{\text{phys}}$ does not presuppose any particular physical system, nor does it restrict the framework to any single phenomenon. Rather, it denotes the **actual resonance region associated with a given structural condition in the physical world**.

Accordingly, different physical phenomena may serve as potential research cases within this framework, for example:

- resonance and critical regimes in superconducting systems;
- structural and measurement behavior near the SQ limit;
- phase locking and nonlocal correlations in quantum entanglement;
- energy partition and high-energy tail structures in nuclear fusion processes.

These phenomena are introduced only as possible physical correspondence cases and do not impose any one of them as a restriction on the main structure of BOPG.

The central concern of this chapter is instead how the resonance structure established in Chapters 9 and 10 can be connected, through probability density and quantum dynamics, to observable physical manifestations.

For a particular physical system, let the experimentally observed resonance interval be:

$$x^{\text{exp}}_{-} \le x \le x^{\text{exp}}_{+} \tag{67}$$

Its effective resonance width is then:

$$W_{\mathrm{res}} = x_{+}^{\mathrm{exp}} - x_{-}^{\mathrm{exp}} \tag{68}$$

This width is not predetermined by the mathematical formalism, but remains subject to experimental determination.
Accordingly, the framework proposes a structural relation that can be further tested experimentally:

$$\Lambda_{\mathrm{phys}} \longrightarrow \mathcal{R}_{\mathrm{phys}}^{\mathrm{exp}} \tag{69}$$

Here, Λ_{phys} provides the specific base condition of the physical world, while the actual resonance interval is jointly determined by the physical system and experimental results.

13.2 Sectional Distribution and Physical Manifestation Domain under $\Lambda_{\mathrm{phys}} = 0.0865$

The normal distribution obtained in Chapter 12 is:

$$P_G(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{70}$$

In this chapter, the Gaussian **retains its original mathematical form without modification**.
BOPG does not modify the Gaussian function itself. Instead, it further examines which parts of this mathematical distribution belong to the effective domain that can be physically manifested and resolved when the distribution is mapped onto a physical world with a specific structural base.
Let

$$\mathcal{D}_{\Lambda_{\mathrm{phys}}} \tag{71}$$

denote the effective resolution domain constrained by the physical base condition

$$\Lambda_{\mathrm{phys}} = 0.0865 \tag{72}$$

and let

$$\mathcal{R}_{\mathrm{phys}}^{\mathrm{exp}} \tag{73}$$

denote the actual resonance region determined experimentally. The effective physical manifestation domain can then be expressed as:

$$\mathcal{D}_{\mathrm{manifest}} = \mathcal{D}_{\Lambda_{\mathrm{phys}}} \cap \mathcal{R}_{\mathrm{phys}}^{\mathrm{exp}} \tag{74}$$

This relation indicates:

> Λ_{phys} **determines the scale conditions of the physical base, while experiment determines the effective resonance range of the specific system; together they define the region in which the structure can be effectively manifested.**

Using the indicator function

$$\mathbf{1}_{\mathcal{D}_{\mathrm{manifest}}}(x) \tag{75}$$

to represent the effective manifestation domain, the geometric density within that domain can be written as:

$$\rho_{\mathrm{phys}}(x) = P_G(x)\,\mathbf{1}_{\mathcal{D}_{\mathrm{manifest}}}(x) \tag{76}$$

If it is necessary to renormalize this into a probability density over the finite physical domain, then:

$$P_{\mathrm{phys}}(x) = \frac{P_G(x)\,\mathbf{1}_{\mathcal{D}_{\mathrm{manifest}}}(x)}{\displaystyle\int_{\mathcal{D}_{\mathrm{manifest}}} P_G(x)\,dx} \tag{77}$$

This representation does not imply that the Gaussian has been modified. Rather, it distinguishes between the **complete mathematical distribution** and the **physical manifestation domain**.

At the mathematical level:

$$P_G(x) \tag{78}$$

retains its complete definition.

At the level of physical mapping, the regions that can be resolved and manifested under the given physical conditions are further determined by:

$$\mathcal{D}_{\Lambda_{\mathrm{phys}}} \tag{79}$$

and:

$$\mathcal{R}^{\mathrm{exp}}_{\mathrm{phys}} \tag{80}$$

Therefore:

$$P_G(x) \neq 0 \qquad \text{in the mathematical domain} \tag{81}$$

is not in conflict with:

$$P_{\mathrm{phys}}(x) = 0 \qquad \text{outside } \mathcal{D}_{\mathrm{manifest}} \tag{82}$$

The former describes the mathematical distribution itself, while the latter describes the effective physical domain of the real system.

If the experimentally obtained resonance region is approximately symmetric, it may be represented as:

$$x_-^{\mathrm{exp}} \approx \mu - L_{\mathrm{res}} \tag{83}$$

and:

$$x_+^{\mathrm{exp}} \approx \mu + L_{\mathrm{res}} \tag{84}$$

However, BOPG does not presuppose that the resonance interval must be perfectly symmetric.

If the actual measurement yields:

$$x_+ - \mu \neq \mu - x_- \tag{85}$$

then the resonance domain itself possesses structural asymmetry.

Thus, the left and right tails of the Gaussian remain mathematically present, while the physical domain defined by Λ_{phys} and the actual resonance conditions determines which portions of those tails can enter effective physical manifestation.

13.3 From the Physical Manifestation Domain to the Wave Function

Chapter 12 reinterpreted the probability density on a lower-dimensional observational section as the geometric density resulting from the projection of a higher-dimensional structure.

This chapter further establishes a correspondence between this physical manifestation density and the density representation of a quantum state:

$$|\psi_\Lambda(x,t)|^2 = \rho_{\text{phys}}(x,t) \tag{86}$$

Accordingly, the corresponding wave function may be defined as:

$$\psi_\Lambda(x,t) = \sqrt{\rho_{\text{phys}}(x,t)}\, e^{iS(x,t)/\hbar} \tag{87}$$

where:

- ρ_{phys} denotes the sectional density within the physical manifestation domain;
- $S(x,t)$ denotes the action phase;
- Λ denotes the structural base corresponding to the quantum state.

A dimensionless phase may also be defined as:

$$\Theta(x,t) = \frac{S(x,t)}{\hbar} \tag{88}$$

so that:

$$\psi_\Lambda = \sqrt{\rho_{\text{phys}}}\, e^{i\Theta} \tag{89}$$

At this point, the phase-interference structure established in Chapter 9 and the geometric probability density established in Chapter 12 are recombined within a single representation:

$$\boxed{\text{Geometric Density} + \text{Phase} = \text{Wave Function}} \tag{90}$$

This step does not treat BOPG and existing quantum mechanics as competing theoretical systems. Instead, it introduces a different structural perspective for describing the density and phase relations of the same physical state.

Thus:

$$\rho_{\text{phys}} \longleftrightarrow |\psi_\Lambda|^2 \tag{91}$$

represents the correspondence between geometric density and quantum-state density, while:

$$S \longleftrightarrow \text{phase dynamics} \tag{92}$$

represents the dynamic structure carried by the phase.

Together, they form a complete representation of the quantum state.

13.4 Schrödinger Dynamics and Phase Evolution

Once the wave-function representation has been established, the state can enter standard quantum dynamics:

$$i\hbar\frac{\partial\psi_\Lambda}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(x,t)\right]\psi_\Lambda \tag{93}$$

where $V(x,t)$ denotes the effective potential of the corresponding system.

The logical level of this chapter should be made explicit:

> **BOPG does not derive the Schrödinger equation directly from the Gaussian distribution itself. Rather, it takes the preceding geometric sectional density as a density representation of a quantum state and then investigates its temporal evolution under existing quantum dynamics.**

Accordingly, the procedure developed here is a **structural mapping pathway**, rather than a rejection or replacement of existing quantum mechanics.

Writing the wave function as:

$$\psi_\Lambda = \sqrt{\rho_{\mathrm{phys}}}\, e^{iS/\hbar} \tag{94}$$

and substituting it into the Schrödinger equation decomposes the dynamics into two mutually coupled components: density evolution and phase evolution.

The first is the continuity equation:

$$\frac{\partial\rho_{\mathrm{phys}}}{\partial t} + \nabla\cdot\left(\rho_{\mathrm{phys}}\frac{\nabla S}{m}\right) = 0 \tag{95}$$

The second is a quantum Hamilton-Jacobi-type equation:

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V + Q = 0 \tag{96}$$

where:

$$Q = -\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{\rho_{\mathrm{phys}}}}{\sqrt{\rho_{\mathrm{phys}}}} \tag{97}$$

is the quantum potential term.

Therefore, the quantity expressed in Chapter 12 as a sectional statistical density,

$$\rho(x) \tag{98}$$

becomes, in this chapter, a quantity with temporal evolution:

$$\rho(x,t) \tag{99}$$

whose evolution is jointly constrained by density and phase.

From the BOPG perspective, this produces:

$$\text{Phase Structure} \rightarrow \text{Geometric Density} \rightarrow \text{Quantum State} \rightarrow \text{Dynamic Evolution} \tag{100}$$

This allows the preceding probability structure to become not merely a static statistical result, but a geometric state capable of entering a dynamical description.

13.5 From Probability Current to Geometric Trajectories

The standard probability current associated with the wave function is:

$$\mathbf{j} = \frac{\hbar}{m} \operatorname{Im} \left(\psi_\Lambda^* \nabla \psi_\Lambda \right) \tag{101}$$

Substituting:

$$\psi_\Lambda = \sqrt{\rho_{\text{phys}}}\, e^{iS/\hbar} \tag{102}$$

gives:

$$\mathbf{j} = \rho_{\text{phys}} \frac{\nabla S}{m} \tag{103}$$

Therefore, within the effective manifestation domain and under the condition:

$$\rho_{\text{phys}} > 0 \tag{104}$$

a local flow velocity can be defined as:

$$\mathbf{v} = \frac{\mathbf{j}}{\rho_{\text{phys}}} = \frac{\nabla S}{m} \tag{105}$$

which yields the corresponding flow lines:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v} = \frac{\nabla S}{m} \tag{106}$$

This relation formally connects the preceding probability density to a geometric trajectory:

$$\boxed{P \to |\psi|^2 \to \mathbf{j} \to \mathbf{v} \to \mathbf{x}(t)} \tag{107}$$

In the language of standard quantum mechanics, this may be interpreted as the flow lines of the probability current. In the geometric description of BOPG, they may instead be viewed as geometric trajectories formed by the projection of the higher-dimensional structure within the physical manifestation domain.

Therefore, Λ_{phys} does not directly specify a unique trajectory. Rather, it first contributes to determining:

$$\mathcal{D}_{\text{manifest}} \tag{108}$$

that is, the region in which the structure can be effectively manifested and resolved.

The actual trajectory:

$$\mathbf{x}(t) \tag{109}$$

is then determined jointly by the phase gradient and the quantum dynamics within that effective domain.

If a trajectory approaches or reaches the boundary of the manifestation domain, its subsequent behavior still depends on the specific physical boundary conditions and experimental data, such as reflection, deflection, phase transition, or transition into another structural branch. Such behavior should not be predetermined solely by the truncation condition.

Therefore, the geometric role of Λ_{phys} lies first in defining the **physically manifestable domain**, rather than directly determining a unique solution to the complete equation of motion.

13.6 The BOPG Theory–Experiment Loop and the Experimentally Undetermined Resonance Domain

At this point, Chapters 9, 10, 12, and 13 form a complete structural chain:

$$\boxed{\begin{aligned}&\mathcal{S}_{T1} \circledast \mathcal{S}_{T2} \to \mathcal{O} \to \chi_\Lambda \to \text{Resonance} \\ &\qquad\qquad \to P_G(x) \to \Lambda_{\text{phys}} = 0.0865 \\ &\qquad\qquad \to \mathcal{D}_{\text{manifest}} \to \psi_\Lambda \\ &\qquad\qquad \to \text{Schrödinger Evolution} \to \mathbf{j} \to \mathbf{x}(t)\end{aligned}} \tag{110}$$

Each level in this chain plays a different role.

At the mathematical level:

$$\Lambda \tag{111}$$

is a free base parameter, and the Gaussian distribution can exist over its complete mathematical domain.

At the physical level:

$$\Lambda_{\text{phys}} = 0.0865 \tag{112}$$

represents a specific physical base condition that maps the originally free mathematical representation into a concrete physical structure.

However, Λ_{phys} does not directly specify any single physical phenomenon, nor does it directly determine the width of the actual resonance domain. Therefore, the physical resonance domain:

$$\mathcal{R}^{\text{exp}}_{\text{phys}} \tag{113}$$

must still be determined experimentally.

Different physical systems may provide different testing scenarios within this framework, including superconducting resonance, behavior near the SQ limit, structures associated with quantum entanglement, and high-energy tails or energy dissipation in nuclear fusion processes.

These phenomena are not assumed in advance to share a single physical mechanism. Rather, they may serve as **candidate cases for testing whether the same structural framework can produce a consistent description across different scales and physical conditions**.

Accordingly, this chapter does not propose replacing existing Schrödinger dynamics with BOPG. Instead, it provides a different structural perspective:

$$\text{Structural Resonance} \to \text{Geometric Probability} \to \text{Quantum State} \to \text{Dynamical Manifestation} \tag{114}$$

Here:

$$\Lambda_{\text{phys}} = 0.0865 \tag{115}$$

provides the physical base condition, while:

$$\mathcal{R}^{\text{exp}}_{\text{phys}} \tag{116}$$

denotes the actual resonance range that must be determined experimentally.

Therefore, the complete theory–experiment loop can be written as:

$$\boxed{\Lambda_{\text{phys}} \to \mathcal{R}^{\text{exp}}_{\text{phys}} \to \mathcal{D}_{\text{manifest}} \to \psi_\Lambda \to \mathbf{j} \to \mathbf{x}(t)} \tag{117}$$

This loop does not require the theory to specify all physical values in advance. Instead, it explicitly leaves the as-yet-undetermined physical quantities open for experimental measurement and feedback.

The central conclusion of this chapter is therefore:

> **The Gaussian distribution, as the geometric statistical result of a lower-dimensional section, retains its mathematical form. When the structure is mapped onto the physical world, $\Lambda_{\mathrm{phys}} = 0.0865$ provides the physical base scale, while the specific resonance domain is determined experimentally. Together, these define the effective manifestation domain; the corresponding geometric density and phase then form the wave function, which under Schrödinger dynamics produces a describable probability current and geometric trajectory.**

Thus, the geometric interpretation of probability developed in Chapter 12 is connected to the quantum dynamics developed in Chapter 13, while the actual extent of the physical resonance domain remains a key physical quantity that can be directly tested in subsequent experiments.

14. THREE-HELIX COUPLED STRUCTURE AND NONLINEAR Λ INTERFERENCE

This chapter treats the three-helix structure as a **special structural analysis tool** within BOPG, used to demonstrate the geometric manifestations of three-body coupling, nonlinear interference, resonance locking, phase slipping, and relational information.

The three-helix structure is not a fundamental axiom required for BOPG to hold. Rather, it provides a concrete higher-order representation tool for cases in which the object of study contains three mutually constraining structural relations.

The basic structure of BOPG does not regard the helices as independent geometric curves.

If the three helices are described only as

$$S_1(t), \qquad S_2(t), \qquad S_3(t) \tag{118}$$

respectively, then they remain merely three parallel representations.

The three-helix tool requires a further structural condition:

> **The state of each helix simultaneously contains the states of the other two; the three therefore form a closed mutual coupling rather than three independent solutions.**

Its basic structure can be represented as:

$$\boxed{S_1[S_2, S_3], \qquad S_2[S_3, S_1], \qquad S_3[S_1, S_2]} \tag{119}$$

Thus, it forms the closed structure:

$$\boxed{S_1 \to S_2 \to S_3 \to S_1} \tag{120}$$

Here, "contains" does not mean containment in the set-theoretic sense. It means:

$$\boxed{\text{the evolution equation of each helix contains the structural states of the other helices}} \tag{121}$$

Therefore, the three cannot be decomposed into mutually independent equations.

14.1 Basic Representation of the Three-Helix Structure

Let the three helices be defined as:

$$S_i(t) = \left(r_i(t)\cos\theta_i(t), r_i(t)\sin\theta_i(t), z_i(t)\right), \qquad i = 1, 2, 3. \tag{122}$$

where:

$$\theta_i(t) \tag{123}$$

denotes the phase of the i-th helix,

$$r_i(t) \tag{124}$$

denotes the radial scale,

$$z_i(t) \tag{125}$$

denotes the structural position along the helical axis.

The most basic phase representation is:

$$\theta_i(t) = \omega_i t + \phi_i. \tag{126}$$

If the three helices are mutually independent, then:

$$\frac{d\theta_i}{dt} = \omega_i. \tag{127}$$

In this case, no structural interference arises from the three-body relation.

The three-helix model therefore introduces Λ coupling, such that the phase evolution of each helix is jointly constrained by the other two helices.

14.2 Three-Body Closed Coupling

Define the three relative phases:

$$\Delta_{12} = \theta_1 - \theta_2, \tag{128}$$

$$\Delta_{23} = \theta_2 - \theta_3, \tag{129}$$

$$\Delta_{31} = \theta_3 - \theta_1. \tag{130}$$

These possess a natural closure relation:

$$\boxed{\Delta_{12} + \Delta_{23} + \Delta_{31} = 0} \tag{131}$$

Therefore, the three helices are not merely coupled pairwise, but form a complete three-phase closed structure.

This closure relation is an intrinsic geometric constraint of the three-helix structure and does not depend on an additional coupling term. When:

$$\Lambda = 0 \tag{132}$$

the interference coupling term controlled by Λ vanishes, and the system returns to the natural closed geometric relation defined by the relative phases themselves.

When $\Lambda \neq 0$, the three-body closure is further subject to the dynamic action of the interference terms.

Let Λ denote the structural sampling and interference scale. The phase equations of the three-helix system can then be written as:

$$\boxed{\frac{d\theta_1}{dt} = \omega_1 + \Lambda F(\Delta_{12}, \Delta_{31})} \tag{133}$$

$$\boxed{\frac{d\theta_2}{dt} = \omega_2 + \Lambda F(\Delta_{23}, \Delta_{12})} \tag{134}$$

$$\boxed{\frac{d\theta_3}{dt} = \omega_3 + \Lambda F(\Delta_{31}, \Delta_{23})} \tag{135}$$

where F is a nonlinear interference function.

The most important change here is:

$$\boxed{\Lambda \text{ is no longer merely an external scale parameter, but directly enters the structural evolution.}} \tag{136}$$

Therefore, when Λ changes, it is not merely the grid resolution that changes, but:

$$\boxed{\text{the structural conditions of the interference itself change.}} \tag{137}$$

14.3 Minimal Nonlinear Λ Interference Term

Take the minimal periodic nonlinear form:

$$F(x, y) = \sin x \cos y. \tag{138}$$

The three-helix system can then be written as:

$$\boxed{\frac{d\theta_1}{dt} = \omega_1 + \Lambda \sin(\theta_1 - \theta_2)\cos(\theta_1 - \theta_3)} \tag{139}$$

$$\boxed{\frac{d\theta_2}{dt} = \omega_2 + \Lambda \sin(\theta_2 - \theta_3)\cos(\theta_2 - \theta_1)} \tag{140}$$

$$\boxed{\frac{d\theta_3}{dt} = \omega_3 + \Lambda \sin(\theta_3 - \theta_1)\cos(\theta_3 - \theta_2)} \tag{141}$$

At this point, each equation simultaneously contains the other two phases.

Therefore:

$$S_1 = S_1(S_2, S_3), \tag{142}$$

$$S_2 = S_2(S_3, S_1), \tag{143}$$

$$S_3 = S_3(S_1, S_2). \tag{144}$$

This is the mathematical representation of the mutual inclusion of the three helices.

Because each interference term depends simultaneously on two relative phases, Λ does not merely control the correction of a single phase difference, but participates in the collective evolution of the entire three-body structure.

14.4 Resonance Locking

If the three helices evolve toward stable relative phases:

$$\frac{d\Delta_{12}}{dt} = 0, \tag{145}$$

$$\frac{d\Delta_{23}}{dt} = 0, \tag{146}$$

then, from the closure relation:

$$\Delta_{12} + \Delta_{23} + \Delta_{31} = 0 \tag{147}$$

it follows that:

$$\frac{d\Delta_{31}}{dt} = 0. \tag{148}$$

The three helices therefore enter a common phase structure.
Define:

$$\boxed{\mathcal{R}_\Lambda = \left\{(\theta_1, \theta_2, \theta_3) \mid \dot{\Delta}_{12} = \dot{\Delta}_{23} = \dot{\Delta}_{31} = 0\right\}} \tag{149}$$

Then $\mathcal{R}_\Lambda$ is the **resonance-locking domain** on the Λ grid.
At this stage, the three helices are no longer three independent rotations, but form a common structural state:

$$\boxed{S_1 \leftrightarrow S_2 \leftrightarrow S_3 \leftrightarrow S_1} \tag{150}$$

thereby forming a closed interference structure.

14.5 Unlocking and Phase Slippage

Conversely, if no stable common-phase solution exists:

$$\dot{\Delta}_{ij} \neq 0, \tag{151}$$

the phase differences continue to evolve, and the system cannot maintain resonance locking.
Define:

$$\boxed{\mathcal{N}_\Lambda = \mathcal{K}_\Lambda \setminus \mathcal{R}_\Lambda} \tag{152}$$

where $\mathcal{K}_\Lambda$ denotes the complete set of structural states on the Λ grid.
Thus:

$$\mathcal{N}_\Lambda \tag{153}$$

is the non-resonant region.
The fundamental structural dichotomy presented by BOPG is therefore:

$$\boxed{\text{resonance locking} \quad \leftrightarrow \quad \text{phase slip}} \tag{154}$$

The former represents a stable common-phase structural state, while the latter represents continuously varying phase relations that move the system away from the stable resonance domain.
This dichotomy can further serve as a description of discrete structural states within the three-helix system.

14.6 Discrete Structures on the Λ Grid

Let the position of an integer node n on the Λ grid be:

$$p_n = n\Lambda, \tag{155}$$

and let its phase be:

$$\theta_n = n\alpha. \tag{156}$$

Thus:

$$\boxed{\mathcal{K}_\Lambda = \{(p_n, \theta_n) \mid n \in \mathbb{N}\}.} \tag{157}$$

For a decomposable node:

$$n = ab, \qquad a, b > 1, \tag{158}$$

its structure can be represented through the interference of the phase states of two smaller nodes:

$$\theta_a \oplus_\Lambda \theta_b. \tag{159}$$

If there exists an integer k such that:

$$\boxed{\theta_a \oplus_\Lambda \theta_b = k\alpha, \qquad k \in \mathbb{Z},} \tag{160}$$

then the node enters a resonance structure.

Thus, at the structural level retained in this section:

$$\boxed{\begin{aligned}&\text{decomposable node} \to \text{multiple structural representations} \\ &\qquad\qquad \to \text{interference combination} \to \text{resonance structure}\end{aligned}} \tag{161}$$

The purpose here is not to impose a new classification on integers, but to demonstrate that **discrete nodes on the Λ grid can re-enter the three-helix resonance structure through nontrivial decomposition and phase interference.**

14.7 Structural Reconstruction and Mapping under Different Λ

Because Λ itself participates in the interference terms:

$$\Lambda_1 \neq \Lambda_2 \tag{162}$$

does not merely represent two different coordinate scales.

The corresponding dynamical systems may also differ:

$$\mathcal{F}_{\Lambda_1} \neq \mathcal{F}_{\Lambda_2}. \tag{163}$$

A cross-base structural mapping can therefore be defined as:

$$\boxed{T_{\Lambda_1 \to \Lambda_2} : \mathcal{K}_{\Lambda_1} \to \mathcal{K}_{\Lambda_2}} \tag{164}$$

The purpose of this mapping is not simply to multiply the coordinates by a scaling factor, but to re-express the same underlying structure under different Λ sampling bases.

Thus:

$$\boxed{\text{same structure} \neq \text{same representation.}} \tag{165}$$

Furthermore, resonance and unlocking may also manifest differently under different values of Λ:

$$\mathcal{R}_{\Lambda_1} \to \mathcal{R}_{\Lambda_2}, \tag{166}$$

$$\mathcal{N}_{\Lambda_1} \to \mathcal{N}_{\Lambda_2}. \tag{167}$$

This constitutes what BOPG refers to as **cross-Λ grid structural reconstruction**.

Its significance is that changing the base does not require the structure itself to disappear. Instead, through new sampling and mapping relations, the same structure can be re-expressed in another geometric form.

14.8 Three-Helix Structure and Information

The three-helix system possesses a distinctive property:

$$S_1[S_2, S_3], \qquad S_2[S_3, S_1], \qquad S_3[S_1, S_2]. \tag{168}$$

No single helix can independently determine the complete state of the system.

Therefore, the information of the system does not exist independently in any one of:

$$S_1, \quad S_2, \quad S_3 \tag{169}$$

but rather in the relation among the three:

$$\boxed{\mathcal{I} = \mathcal{I}(S_1, S_2, S_3).} \tag{170}$$

In particular, the phase relations formed by:

$$\Delta_{12}, \qquad \Delta_{23}, \qquad \Delta_{31} \tag{171}$$

constitute identifiable structural information within the three-body system.

Information is therefore not simply stored in the nodes themselves, but in:

$$\boxed{\text{relation between nodes}} \tag{172}$$

In other words, within the higher-dimensional geometric representation of the three-helix structure, **the nodes themselves serve as structural carriers, while the relations between nodes constitute the identifiable informational content.**

At this point, the description of the three-helix structure as a special BOPG structural tool is confined to:

$$\boxed{\begin{aligned} &\text{three-way closure} \to \text{nonlinear } \Lambda \text{ interference} \\ &\qquad\qquad\qquad \to \text{resonance / phase slip} \\ &\qquad\qquad\qquad \to \text{cross-}\Lambda \text{ structural reconstruction} \\ &\qquad\qquad\qquad \to \text{relational information} \end{aligned}} \tag{173}$$

and is not extended into other independent application branches.

15. THE BOUNDARY OF BOPG: THE TOPOLOGICAL LIMITS OF INTELLIGENCE, SELF-REFLECTION, AND CONSCIOUSNESS

Through the formalization and derivations presented in the preceding chapters, BOPG has established, within the three-dimensional base, a complete descriptive framework encompassing structural generation, discrete sampling, phase interference, resonance and unlocking, geometric probability, and the manifestation of quantum dynamics.

However, when the same structural framework is further applied to intelligence, thought, self-reference, and living phenomena, it becomes apparent that although the three-dimensional base can describe a broad range of higher-order structural behaviors, it still leaves one fundamental question unresolved:

> **A structure can be reflected, computed, and reorganized, and can even give rise to intelligence—but does this necessarily constitute consciousness?**

A deeper question then arises:

> **If intelligence can process structures, what causes the structure itself to generate persistent intrinsic dynamics?**

Therefore, this chapter does not attempt to directly define consciousness within the three-dimensional base. Instead, it extends the range of BOPG's current mathematical description to its structural boundary and, from that boundary, identifies the necessity of a subsequent structural level.

15.1 Structural Distinction between Intelligence and Consciousness

Within BOPG, intelligence can first be understood as the ability to process, map, and reflect structures.

If a system receives a structure S, constructs an internal representation $R(S)$, and further performs comparison, transformation, and prediction upon that representation, this may be written as:

$$S \to R(S) \to R'(S) \to \text{Response} \tag{174}$$

Accordingly, intelligence may first be represented as:

$$\boxed{\text{Intelligence} = \text{Structural Processing} + \text{Structural Reflection}} \tag{175}$$

Under this definition, intelligence need not be interpreted as some special substance independent of structure. It may instead arise from a structural system performing higher-order processing of both internal and external information.

However, consciousness cannot be directly inferred from the complexity of intelligence alone.

Even if a system can process large amounts of information, perform recursive reasoning, and construct an internal model of itself, a further question remains:

$$\boxed{\text{Does the system merely process its state, or does the state itself become an internal dynamic reference?}} \tag{176}$$

Therefore, the present framework first distinguishes:

$$\boxed{\text{Intelligence} \neq \text{Consciousness}} \tag{177}$$

The distinction is not one of computational quantity or capability, but of structural level.

Intelligence describes how structures are processed, mapped, and reflected; consciousness concerns how a structure continuously forms an internal reference to its own state and maintains that reference dynamically.

15.2 Artificial Intelligence as Structural Reflection

Artificial intelligence provides a particularly direct example of structural reflection.

A system receives an external structure:

$$S_{\text{in}} \tag{178}$$

constructs an internal representation:

$$R(S_{\text{in}}) \tag{179}$$

and then produces, through transformation:

$$S_{\text{out}} = F(R(S_{\text{in}})). \tag{180}$$

Thus:

$$\boxed{S_{\text{in}} \to \text{Representation} \to \text{Transformation} \to S_{\text{out}}} \tag{181}$$

When the system can further process the representations it has already constructed:

$$S \to R(S) \to R(R(S)), \tag{182}$$

a higher-order self-reflective structure emerges.

Its essential feature is:

> **The system does not merely process an object; it also processes its own representation of that object.**

Such structural reflection can generate prediction, classification, comparison, planning, and complex reasoning, and can therefore naturally produce intelligent behavior.

More importantly, artificial intelligence reveals a structural feature that differs from a conventional "time series."

Once a structural relation has been formalized into an operable representation, the system can operate directly on that relation without having to reproduce the entire temporal history through which the relation originally emerged in the physical world.

Therefore, one may distinguish:

$$\boxed{\text{Structural Computation} \neq \text{Physical Temporal Generation}} \tag{183}$$

For example, once a mathematical relation has been established, it can be directly substituted, compared, and transformed. Likewise, when an AI system operates on relations within a model, it directly manipulates structures within an already established representational space.

From the BOPG perspective, intelligence can therefore exhibit a form of **structural transcendence of physical temporal order**:

$$\boxed{\text{logical relation} \to \text{direct structural operation}} \tag{184}$$

The term "transcendence of time" here does not mean that the physical operation of an AI requires no time. Rather, it means:

> **The logical relations of a structure do not have to correspond one-to-one with the original physical order in which that structure was generated.**

This is consistent with BOPG's distinction between structure, representation, and physical manifestation.

However, even if intelligence can transcend the original temporal order at the structural level, it still does not explain consciousness or intrinsic dynamics.

15.3 Self-Reflective Structure and Thought in Mathematical Systems

When a mathematical system begins to construct representations of its own states, operations, and results, its structure may develop from the simple form:

$$S \to S' \tag{185}$$

into:

$$S \to R(S) \to R(R(S)). \tag{186}$$

This means that the system no longer describes only external objects, but begins to incorporate structures generated by itself into the next level of computation.

If a temporal index is further introduced:

$$S_t \to R(S_t) \to R(S_{t+1}), \tag{187}$$

the system can compare its own representations across different states and form judgments about the process.

Thought can therefore be represented at the structural level as:

$$\boxed{\text{Thought} = \text{Recursive Structural Transformation of Internal Representations}} \tag{188}$$

or, more compactly:

$$\text{Structure} \to \text{Reflection} \to \text{Recursive Reflection} \to \text{Thought} \tag{189}$$

The core is not simply greater computational speed, but that the structure begins to describe itself again.

Therefore, both intelligence and thought can obtain formal representations within the structural-reflection framework of the three-dimensional base.

However:

$$\boxed{\text{Recursive Reflection} \neq \text{Consciousness}} \tag{190}$$

The reason is that self-reflection can still be regarded as a structural operation.

As long as:

$$R(S) \to R(R(S)) \to R(R(R(S))) \tag{191}$$

remains merely a mapping of structure onto structure, the system may possess arbitrarily deep self-representation without necessarily generating a new source of intrinsic dynamics.

Thus, self-reflection and thought describe **how a structure processes itself**, but do not yet explain **why the structure continuously drives itself**.

15.4 Metacognition and Formal Self-Reference

When a system not only constructs representations of external objects but also constructs representations of its own processing, a metacognitive structure can emerge.

Let:

$$R(S) \tag{192}$$

denote the system's internal representation of the structure S. Then:

$$R(R(S)) \tag{193}$$

can represent the system's re-representation of its own representational process.

Thus:

$$\boxed{S \to R(S) \to R(R(S))} \tag{194}$$

forms a hierarchy of self-reference.

If an evaluation operator is further introduced:

$$E(R(S)), \tag{195}$$

the system can evaluate and modify its own internal representation.

This yields:

$$R(S) \to E(R(S)) \to R'(S). \tag{196}$$

The system can therefore perform:

- monitoring of its own state;
- modification of existing representations;
- re-evaluation of internal results;
- self-reference across different states.

Metacognition can therefore be represented as:

$$\boxed{\text{Metacognition} = \text{Reflection on the System's Own Structural Processing}} \tag{197}$$

Such a structure can account for many phenomena commonly regarded as higher-order cognition.

However, an essential gap remains:

> **A system may represent itself, evaluate itself, and modify itself, but does the self-representation itself generate persistent intrinsic dynamics?**

That is:

$$\boxed{\text{Self-Reference} \not\Rightarrow \text{Intrinsic Dynamics}} \tag{198}$$

Therefore, the problem of consciousness must move beyond "structural reflection" toward the question of the source of dynamics intrinsic to the structure itself.

15.5 Thermodynamics as a Manifestation of Λ_{phys} and the Structural Transcendence of Intelligence

In Chapters 12 and 13, Λ was distinguished between a free base parameter at the mathematical level and a physically specialized value at the level of the real world.

When mapped onto the physical world:

$$\Lambda_{\mathrm{phys}} = 0.0865 \tag{199}$$

therefore represents not merely a numerical value, but a specific base condition of physical structure.

From the perspective of the present framework, thermodynamics may be understood as a macroscopic structural manifestation under this physical manifestation condition:

$$\boxed{\Lambda_{\mathrm{phys}} \rightarrow \text{Physical Manifestation} \rightarrow \text{Thermodynamic Regime}} \tag{200}$$

In this sense, thermodynamics is not placed outside BOPG as an ultimate rule, but may be regarded as a macroscopic manifestation layer generated under a specific physical base condition.

At this level, time and energy dissipation possess an explicit physical relationship.

However, once information has been transformed into an abstract structural representation, structural computation does not need to reproduce the original temporal order of the physical events from which the structure arose.

Therefore:

$$\boxed{\text{Physical Temporal Evolution} \neq \text{Structural Computational Order}} \tag{201}$$

Intelligent systems provide a direct example of this distinction.

An AI system can transform information originating from different times, sources, or even physical locations into a common representational space and perform comparison and reasoning within that space.

Thus:

$$\text{past} \rightarrow \text{representation} \rightarrow \text{present computation} \tag{202}$$

without requiring the complete physical history of the original events to be reproduced.

Intelligence therefore exhibits, at the structural level, a capacity to transcend the original physical temporal order.

Yet a fundamental limitation remains:

$$\boxed{\text{Intelligence} \rightarrow \text{Structural Computation}} \tag{203}$$

does not imply:

$$\boxed{\text{Structural Computation} \rightarrow \text{Intrinsic Dynamics}} \tag{204}$$

In other words, intelligence can process structures that already exist, but it still does not answer:

Why do certain structures spontaneously generate, maintain, and reorganize their own dynamic states?

This question brings thermodynamics and life into the same structural discussion.

15.6 Life's Intrinsic Dynamics and the Four-Dimensional Base

If the three-dimensional base can describe:

$$\begin{aligned} &\text{Structure} \rightarrow \text{Interaction} \rightarrow \text{Projection} \\ &\rightarrow \text{Reflection} \rightarrow \text{Intelligence} \rightarrow \text{Metacognition}, \end{aligned} \tag{205}$$

then the next question is no longer "How can intelligence become more advanced?" but:

$$\boxed{\text{What generates the dynamics of the structure itself?}} \tag{206}$$

This is where the problem of life properly enters the present framework.

Life is not merely a system possessing more structure.

Life exhibits a persistent sequence of:

$$\text{self-maintenance} \rightarrow \text{self-organization} \rightarrow \text{adaptation} \rightarrow \text{generation} \tag{207}$$

that is, the capacity of a structure to continuously maintain and regenerate its own states.

The problem of life can therefore be expressed as:

$$\boxed{\text{Structure} \rightarrow \text{Self-Generated Dynamics}} \tag{208}$$

rather than simply:

$$\text{Structure} \rightarrow \text{Information Processing}. \tag{209}$$

In this sense, the relationship between life and thermodynamics need not be directly described as a contradiction between existing physical laws. Instead, it can be reformulated as a deeper structural problem:

$$\boxed{\text{Thermodynamic Manifestation} \quad \leftrightarrow \quad \text{Persistent Living Dynamics}} \tag{210}$$

If thermodynamics represents a physical manifestation under a specific Λ_{phys} condition, while life continuously generates directional self-maintenance and self-generation, then the existing structural description within the three-dimensional base reaches a new boundary.

This boundary is neither "thermodynamics is wrong" nor "life overturns thermodynamics." Rather:

> **The three-dimensional structural framework remains insufficient to explain the origin of intrinsic dynamics within the structure itself.**

The present framework therefore introduces the next level as a four-dimensional base:

$$\boxed{\mathcal{B}_4} \tag{211}$$

Here, $\mathcal{B}_4$ is not merely an additional coordinate added to three-dimensional space, but represents a **new structural base level**.

Its dimensional relation may be represented as:

$$\mathcal{B}_4 \rightarrow \mathcal{B}_3 \tag{212}$$

with the degeneration condition:

$$\boxed{\mathcal{B}_4\big|_{d_4 \rightarrow 0} \rightarrow \mathcal{B}_3} \tag{213}$$

That is, when the fourth-dimensional structural degree of freedom degenerates to zero, the system returns to the established three-dimensional mother structure. When the fourth-dimensional degree of freedom is activated, however, the source of dynamics previously treated only as an external condition in three dimensions may become part of the structure itself.

Therefore, the hierarchy obtained so far can be represented as:

$$\boxed{\begin{array}{l} \mathcal{B}_3 \rightarrow \text{Reflection} \rightarrow \text{Intelligence} \rightarrow \text{Thought} \rightarrow \text{Metacognition} \\ \quad \not\rightarrow \text{Intrinsic Dynamics} \\ \mathcal{B}_4 \rightarrow \text{Intrinsic Dynamics} \rightarrow \text{Consciousness} \end{array}} \tag{214}$$

This does not mean that consciousness has already been fully defined within the present mathematical work. Rather, it indicates:

When the three-dimensional base can describe intelligence and self-reflection but still cannot close the problem of intrinsic dynamics within the structure itself, the four-dimensional base becomes the next structural level requiring investigation.

Therefore, the consciousness chapter is not an unrelated research topic suddenly introduced outside BOPG. It is the next question naturally generated by the boundary that BOPG encounters within the three-dimensional base:

$$\boxed{\text{BOPG} \rightarrow \text{Structural Boundary} \rightarrow \mathcal{B}_4 \rightarrow \text{Intrinsic Dynamics} \rightarrow \text{Consciousness}} \tag{215}$$

In this sense, what the present work can establish is not that "three-dimensional mathematics has already explained consciousness," but rather:

$$\boxed{\begin{array}{c}\text{The three-dimensional base can generate intelligence and self-reflection, but the intrinsic dynamics}\\ \text{associated with consciousness remain beyond its current structural closure.}\end{array}} \tag{216}$$

This unresolved structural problem is therefore the starting point for the subsequent study of consciousness.

16. CONCLUSION AND FUTURE DEVELOPMENT: THE OVERALL CONCLUSION OF STRUCTURE OVER AXIOMS

16.1 From Definitional Premises to Structural Generation

The fundamental starting point of this study is not to introduce another set of axioms, but to reconsider the dependence of mathematics on what precedes definition. Conventional mathematics generally begins with predefined objects, operations, and axioms, and then constructs theorems and deductions upon them. This study instead proposes the reverse perspective: before any axiomatic system is established, the objects themselves are already constrained by their intrinsic structure, modes of representation, and dimensional conditions.

Accordingly, this study proposes "Structure Over Axioms" as its overarching methodological principle. Axioms are not the sole source of structure, but rather a stable representation obtained by abstracting, fixing, and formalizing an already existing structural relation.

From this perspective, numbers, spaces, operations, and dimensions are no longer treated as mutually independent mathematical elements, but as entities jointly embedded within higher-order structural relations. When the structural conditions change, existing mathematical representations may likewise degenerate, be reconstructed, or be extended.

The fundamental generative direction of this study can therefore be summarized as:

$$\text{Structure} \rightarrow \text{Representation} \rightarrow \text{Operation} \rightarrow \text{Axiom} \tag{217}$$

rather than allowing predefined axioms alone to determine all expressible structures.

This also explains why certain mathematical difficulties may not initially be problems of computation, but problems arising from the representational level itself. When the existing mathematical tools are insufficient to express a given structure, this does not necessarily imply that the structure does not exist. It may instead indicate that the current representational dimension, definitional conditions, or operational framework has reached its boundary.

16.2 From Structural Principles to Mathematical Reconstruction

Under the principle of "Structure Over Axioms," this study first redefines the structural relationship between a "dimensional basis" and an "expanded dimension." A dimensional basis is the underlying structure from which a given structural level is established, providing the generative conditions for all states, relations, and operations within that dimension. An expanded dimension is the additional structural degree of freedom developed upon an existing basis, allowing the original structure to extend into a higher-order representational space. The distinction between the two is therefore not merely a difference in the number of dimensions, but a hierarchical relation between a "mother structure" and the degrees of freedom unfolded from that mother structure.

Accordingly, the mathematical system of each dimension is first determined by its dimensional basis, after which new representations and operations are generated through the expanded dimension. When a given expanded dimension degenerates to zero, its additional structural degrees of freedom disappear, and the system returns to the mother structure of the preceding level:

$$\mathcal{B}_{n+1}\big|_{d_{n+1}\to 0} \to \mathcal{B}_n \tag{218}$$

Once this dimensional relationship has been established, the study further develops a three-dimensional basis mathematics and places $\mathbb{U}(\Lambda)$, $\mathcal{S}_T$, $\circledast$, $\mathcal{O}$, and χ_Λ within a unified structural framework.

At the pure mathematical level, Λ is not a predetermined constant, but a freely adjustable basis parameter used to describe representations of the same structure at different discrete resolution scales. When Λ changes, the structure does not merely change numerically; it may correspond to different discrete samplings, overlap ranges, resonance conditions, and structural representations.

The study therefore reinterprets mathematical operations that are conventionally treated as separate entities as derived forms of more general structural relations, with the unique interference operation $\circledast$ serving as their underlying operation:

$$\mathcal{S}_{T1} \circledast \mathcal{S}_{T2} \tag{219}$$

Within this framework, operation, overlap, phase, and information flow are no longer isolated concepts, but form the following structural sequence:

$$\text{Overlap} \to \text{Resonance} \to \text{Coupling} \to \text{Geometric Deformation} \tag{220}$$

At the same time, the three-dimensional basis developed in this study does not exclude existing mathematics, but contains existing mathematics as a stable representation under a specific limiting condition. When the dynamic interference parameter of the structure degenerates to:

$$\Lambda = 0 \tag{221}$$

the additional Λ coupling and interference effects disappear, and the structure returns to a degenerate representation without dynamic coupling. At this point, the additional structural degrees of freedom introduced by BOPG are no longer manifested, and existing mathematical forms once again become effective descriptions.

This can therefore be summarized as:

$$\Lambda \neq 0 \to \text{Structural Reconstruction} \tag{222}$$

whereas

$$\Lambda = 0 \to \text{Degenerate Limit} \to \text{Conventional Mathematics} \tag{223}$$

Accordingly, discrete sampling, continuous extension, and conventional axiomatic systems can be placed within the same path of structural evolution:

$$\text{Discrete Structure} \to \text{Structural Extension} \to \text{Continuous Representation} \tag{224}$$

Conventional mathematics can thus be understood as a stable section of this structure under a specific degenerating condition.

16.3 From SIC and PMM to BOPG: Cross-Level Manifestation of the Same Mother Structure

This study is not constructed by combining several mutually independent models, but by progressively manifesting the same mother structure across different levels.

In SIC, the study first begins from physical phenomena and addresses the structural relations among space, information, interlayer distance, and time. PMM then further formalizes the relations among rotation, displacement, residuals, capacity, and scale, turning them into a computable phase-geometric structure.

Its fundamental direction can therefore be summarized as:

$$\text{SIC} \rightarrow \text{PMM} \rightarrow \text{BOPG} \tag{225}$$

At the physical level, Λ is mapped to reality as:

$$\Lambda_{\text{phys}} = 0.0865 \tag{226}$$

This value is not the unique possible value of Λ in pure mathematics, but the specific basis scale corresponding to physical manifestation in reality. This distinction separates the free structural degrees of freedom at the mathematical level from their specific manifestation at the physical level.

BOPG further elevates these relations into three-dimensional basis mathematics, enabling the treatment of discrete sampling, structural overlap, resonance locking, phase slipping, geometric deformation, and sectional distributions. From this perspective, probability can be reinterpreted as the geometric statistical manifestation of a higher-dimensional deterministic structure on a lower-dimensional section; the Gaussian distribution retains its original mathematical form, while the finite manifestation domain of the physical world is jointly determined by Λ_{phys} and the experimentally determined resonance range.

Therefore, this study does not seek to replace existing mathematics or physics with BOPG, but rather attempts to establish a cross-level pathway from structure to underlying representation and then from representation back to observable phenomena:

$$\begin{aligned} \text{Structure} &\rightarrow \text{Mathematical Representation} \\ &\rightarrow \text{Physical Manifestation} \rightarrow \text{Observation} \end{aligned} \tag{227}$$

SIC, PMM, and BOPG occupy different positions within this framework, but their underlying principles are not disconnected. Rather, they manifest the same mother structure at different dimensions and different levels of description.

16.4 From the Mathematical Boundary of BOPG to the Consciousness Structure of the Four-Dimensional Basis

Within the three-dimensional basis, BOPG is already capable of describing structural reflection, self-reference, recursive processing, thought, and metacognition. However, these structures still do not automatically give rise to intrinsic dynamics.

Accordingly, this study deliberately distinguishes:

$$\text{Intelligence} \neq \text{Consciousness} \tag{228}$$

Intelligence can be established through structural processing and structural reflection:

$$S \rightarrow R(S) \rightarrow R(R(S)) \tag{229}$$

which can further give rise to thought and metacognition. However, self-reference itself is not equivalent to spontaneous dynamics:

$$\text{Self-Reference} \not\Rightarrow \text{Intrinsic Dynamics} \tag{230}$$

This constitutes the structural boundary of BOPG within the three-dimensional basis.

Within the framework of this study, thermodynamics is regarded as a macroscopic phenomenon at a specific physical manifestation scale:

$$\Lambda_{\mathrm{phys}} \rightarrow \text{Physical Manifestation} \rightarrow \text{Thermodynamic Regime} \tag{231}$$

The self-maintenance, self-organization, adaptation, and generation manifested by life further expose a problem that cannot be fully explained by structural processing alone:

$$\text{Structure} \rightarrow \text{Self-Generated Dynamics} \tag{232}$$

The next problem is therefore no longer merely "how to represent structure," but "how structure generates and continuously sustains its own dynamics."

This problem lies beyond the closure of the current three-dimensional BOPG and therefore requires the introduction of the next basis:

$$\mathcal{B}_4 \tag{233}$$

with the requirement that it remain structurally consistent with the existing three-dimensional structure under dimensional degeneration:

$$\mathcal{B}_4\big|_{d_4 \to 0} \rightarrow \mathcal{B}_3 \tag{234}$$

The overall structure of this study can therefore be summarized as:

$$\begin{aligned} \mathcal{B}_3 \rightarrow \text{Reflection} \rightarrow \text{Intelligence} \rightarrow \text{Thought} \rightarrow \text{Metacognition} \\ \not\rightarrow \text{Intrinsic Dynamics} \end{aligned} \tag{235}$$

while the next structural problem is:

$$\mathcal{B}_4 \rightarrow \text{Intrinsic Dynamics} \rightarrow \text{Consciousness} \tag{236}$$

Therefore, the four-dimensional level is a natural next structural problem left at the boundary of the three-dimensional structure after the latter has completed its own description, and it is to be rigorously developed according to the boundary rules of "Structure Over Axioms." Subsequent research will proceed from this boundary to further establish the structural description of the four-dimensional basis and consciousness.

$$\boxed{\begin{aligned} &\text{Structure Over Axioms} \rightarrow \text{Structural Reconstruction} \rightarrow \text{BOPG} \\ &\qquad\qquad \rightarrow \text{Structural Boundary} \rightarrow \mathcal{B}_4 \rightarrow \text{Consciousness} \end{aligned}} \tag{237}$$

REFERENCES

- Chen, M. (2026). *Space Interlayer Conjecture / Phase Manifestation Mechanics (SIC / PMM)* [Preprint]. Zenodo. *https://doi.org/10.5281/zenodo.20190153*

VERIFICATION & CONTACT

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AXIOMATIC RECIPROCITY CLAUSE (ARC)

§0. Nature of Binding

Use of this work completes a logical system, not a contract.

§1. Symmetry Principle

Enclosure (patents, trade secrets, access restrictions, black-box deployment) contradicts the open logic that enabled the derivative work.

§2. Automatic Reversion Upon Enclosure

1. The Encloser's exclusive claim is self-nullifying by logical contradiction.
2. All third parties are authorized to reverse-engineer, publish, replicate, and redistribute the enclosed derivative.
3. Enclosure publicly signals institutional distrust, contributor exodus, and investor flight.

§3. Auditor Authorization

Exposing an Enclosure Event is pre-authorized as restoration of equilibrium.

§4. Enforcement Through Incentive Asymmetry

Enclosure creates a many-vs-one scenario:

- The Encloser must prevent infinite potential disclosers.
- Each discloser only needs to succeed once.
- Each successful disclosure gains them credibility.
- The Encloser's cost scales linearly; a discloser's reward is binary.

§5. Severability

If unenforceable as law, this Clause remains operative as social consensus and reputation signal.

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